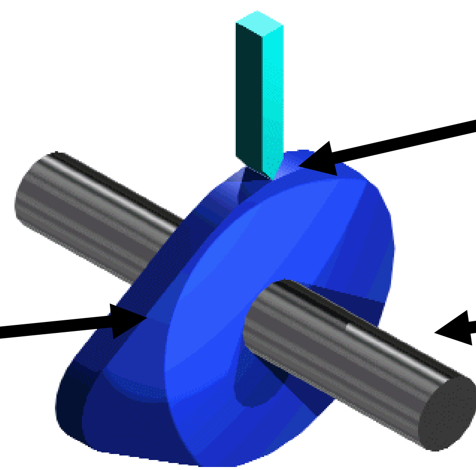
Cam Design (Ch 8)

What is a cam-follower mechanism?

- Cam and follower mechanisms are used to convert rotary motion into reciprocating motion by direct contact.
- The motion created can be simple and regular or complex and irregular.
- As the cam turns, driven by the circular motion, the cam follower traces the surface of the cam transmitting its motion to the required mechanism.
- Disadvantages
  - » High manufacturing cost
  - » Poor wear resistance
  - » Relatively poor high-speed capability

Cam and follower mechanism

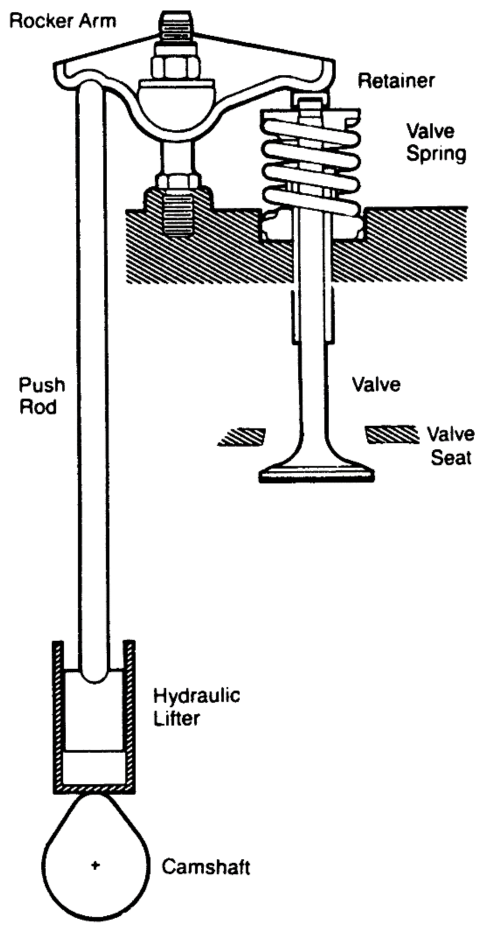


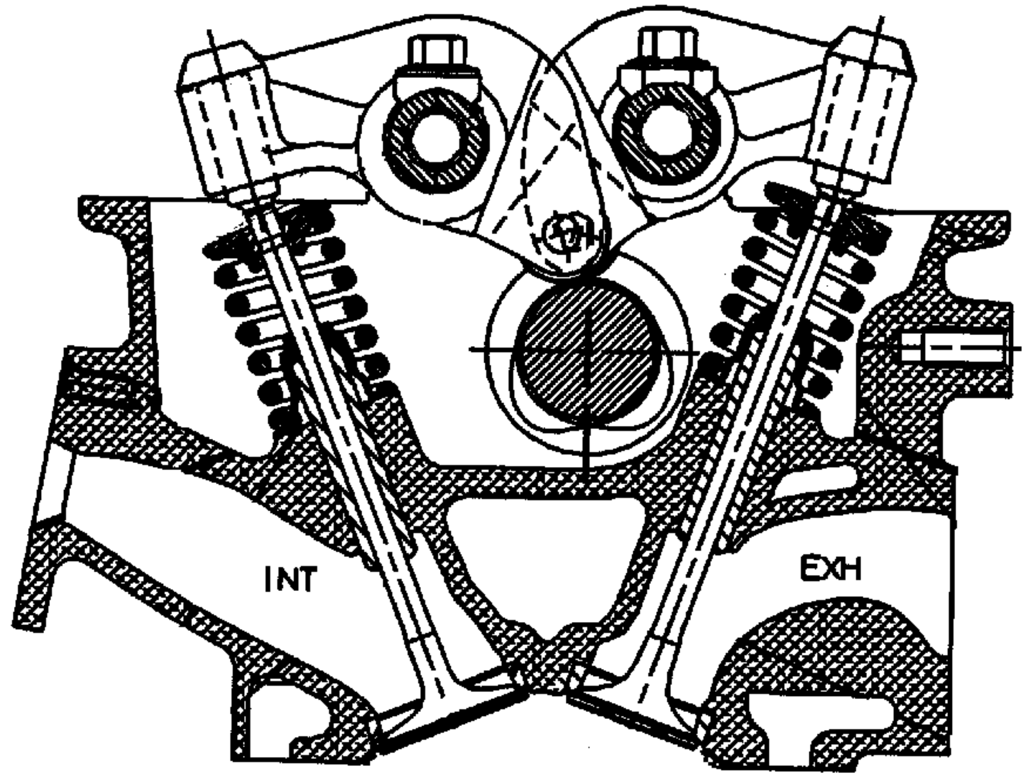
Animation 2 Animation 3

What are Cams Used For?

- Valve actuation in IC engines
- Motion control in machinery
- Force generation
- Precise positioning
- Event timing

Valve Trains



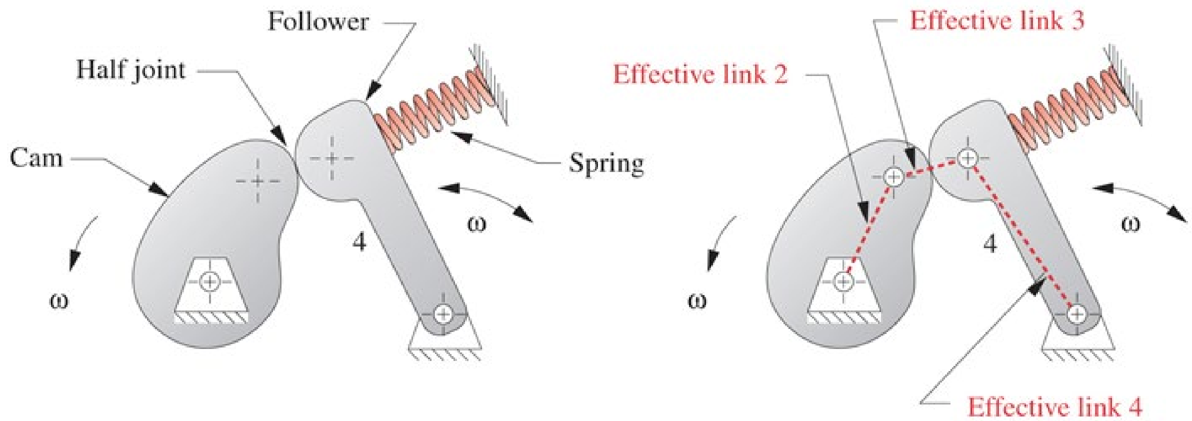


Overhead Valve

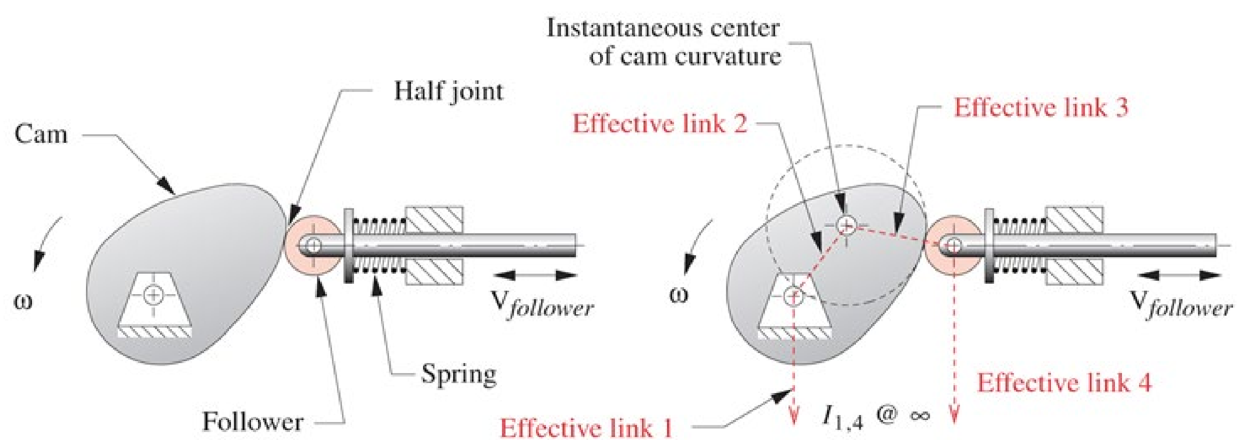
Overhead Camshaft

Cam classification

- **Type of follower motion**



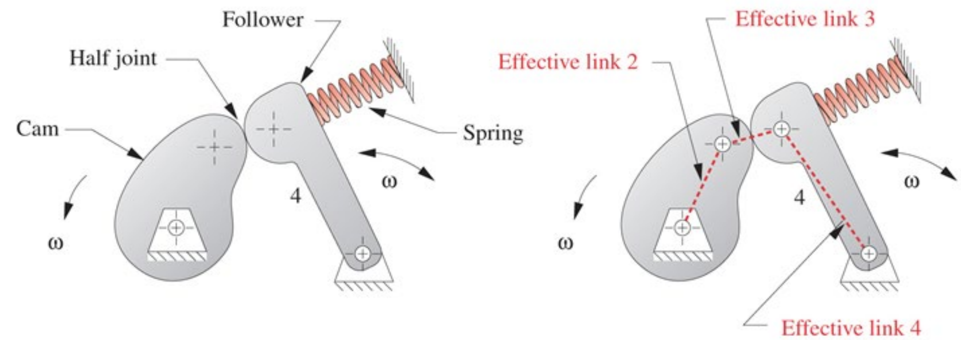
(*a*) An oscillating cam-follower has an effective pin-jointed fourbar equivalent

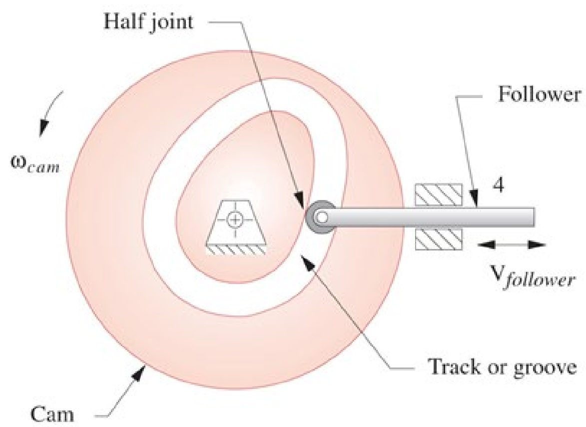


(*b*) A translating cam-follower has an effective fourbar slider-crank equivalent

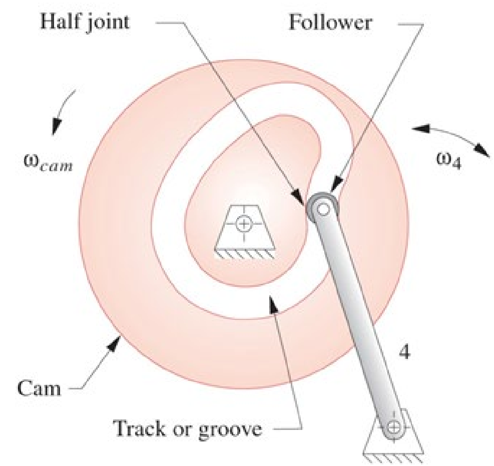
- **Type of joint closure**

Force closure



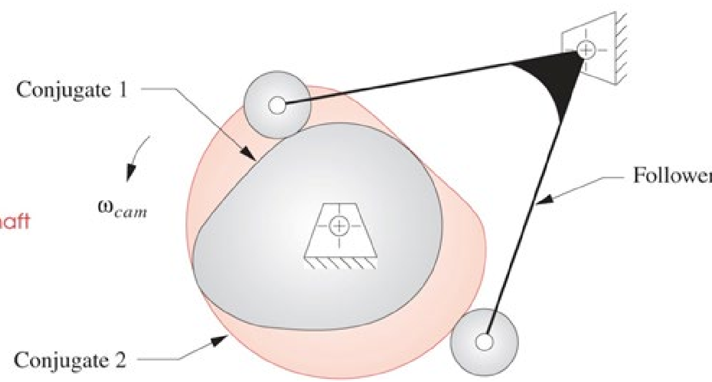


(*a*) Form-closed cam with translating follower



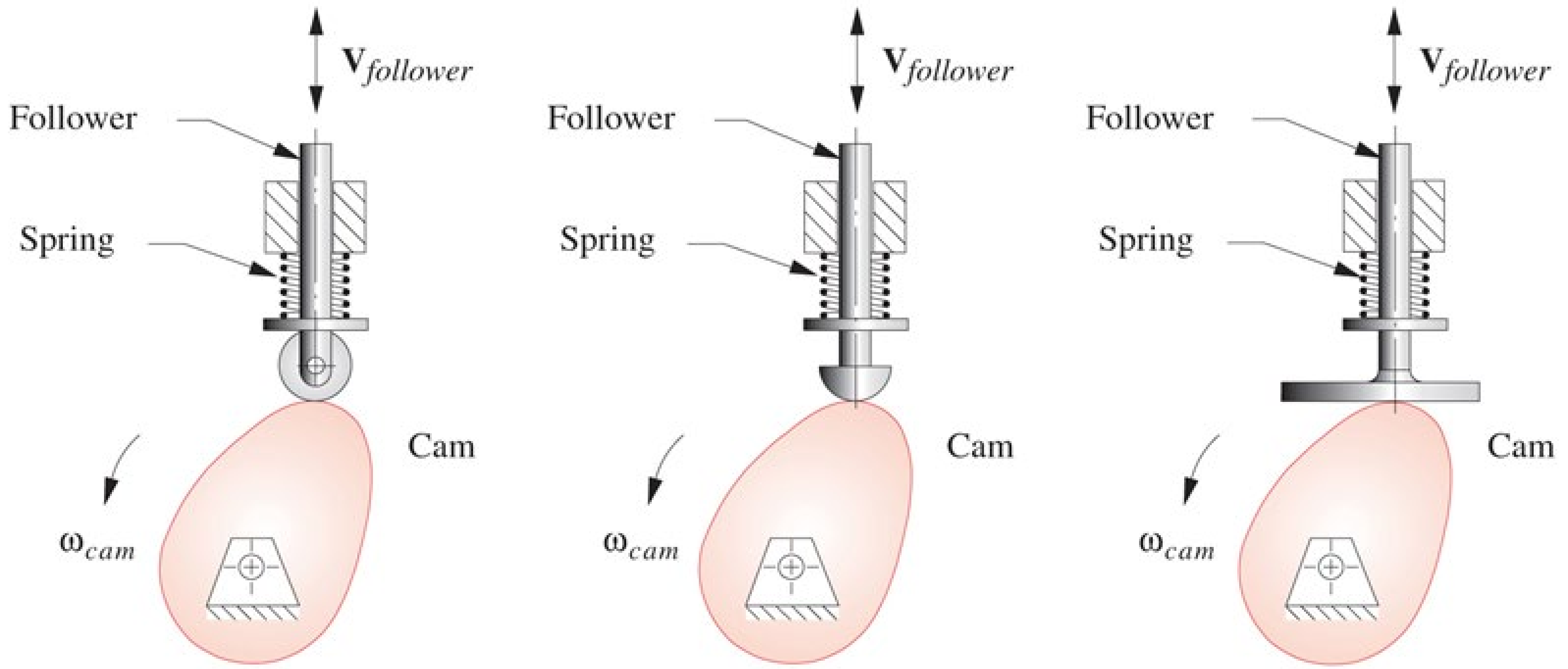
(*b*) Form-closed cam with oscillating follower

Form closure



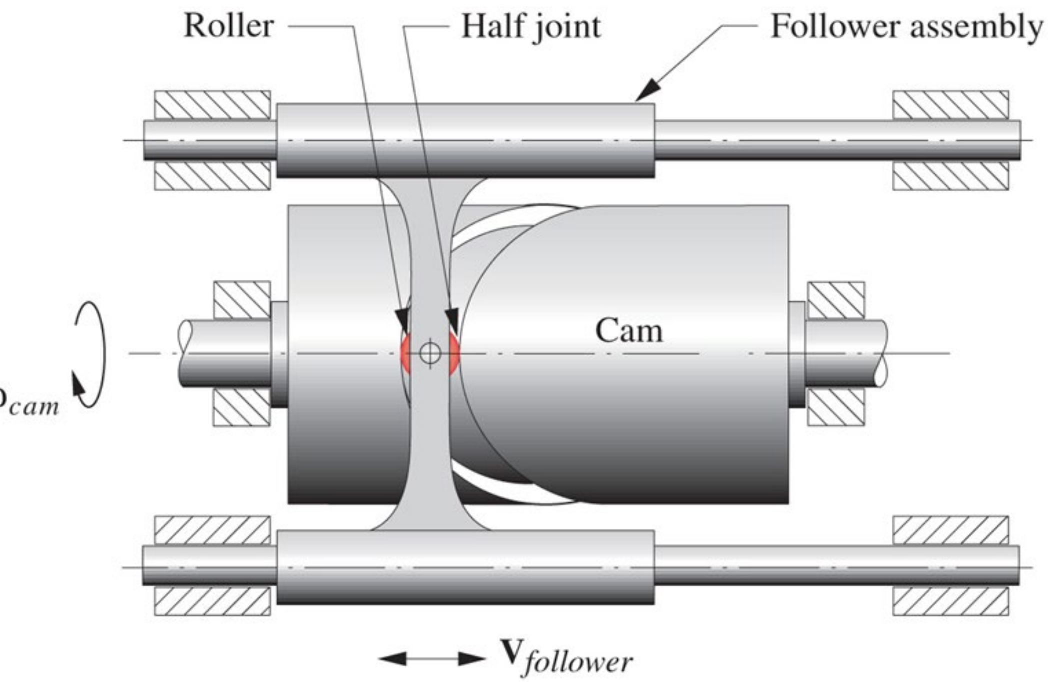
(*c*) Conjugate cams on common shaft

- **Type of follower**

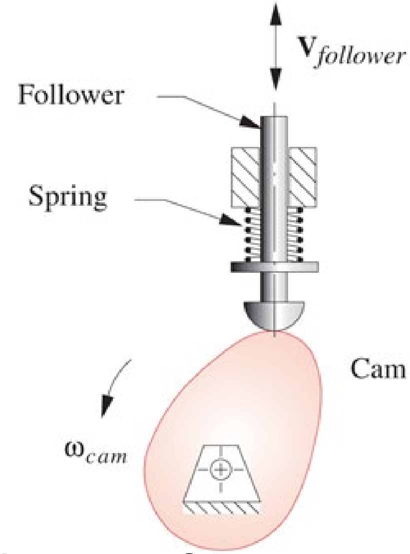


(*a*) Roller follower (*b*) Mushroom follower (*c*) Flat-faced follower

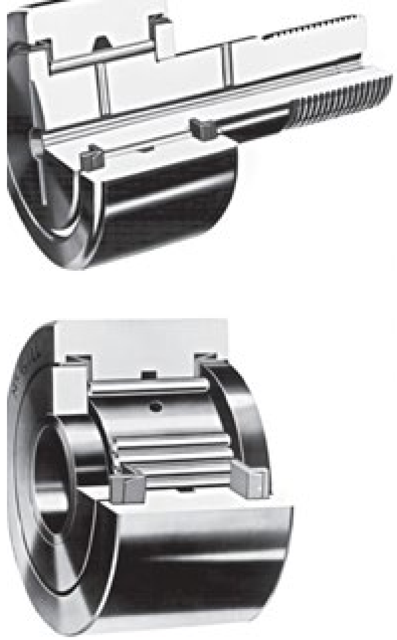
- **Type of cam**



Axial cam: follower moves parallel to the axis of cam; *cylindrical* or *barrel* cam,



Radial cam: follower moves in radial direction



(*a*) Commercial roller followers

*Courtesy of McGill Manufacturing Co.
South Bend, IN*

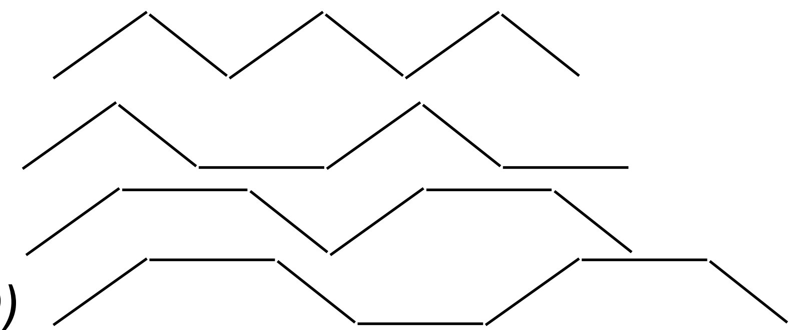
(*b*) Commercial cams of various types

*Courtesy of The Ferguson Co.
St. Louis, MO*

(*c*) Three-dimensional cams

- **Type of motion constraints**
  - *Critical extreme position* (CEP): the start and finish position of the follower are specified without any constraints on the path between the extreme positions
  - *Critical path motion* (CPM): the path motion and/or one or more of its derivatives are defined over all or part of the interval of motion.
- **Type of motion program**
  - *Rise-fall*(RF)
  - *Rise-fall-dwell (RFD)*
  - *Rise-dwell-fall (RDF)*
  - *Rise-dwell-fall-dwell (RDFD)*

    or *Dwell-rise-dwell-fall (DRDF)*

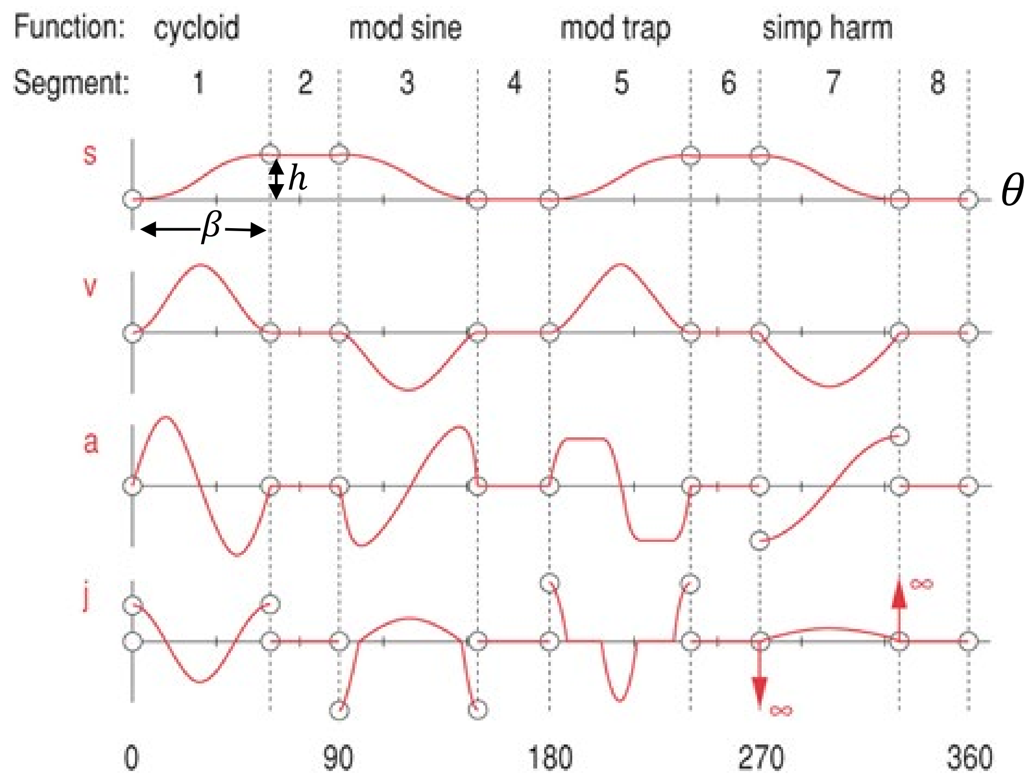


SVAJ diagrams

- S V A J：displacement, velocity, acceleration and jerk

Segment Number	Function Used	Start Angle	End Angle	Delta Angle
1	Cycloid rise	0	60	60
2	Dwell	60	90	30
3	ModSine fall	90	150	60
4	Dwell	150	180	30
5	ModTrap rise	180	240	60
6	Dwell	240	270	30
7	SimpHarm fall	270	330	60
8	Dwell	330	360	30

(a) Cam program specifications



(b) Plots of cam-follower's *s v a j* diagrams

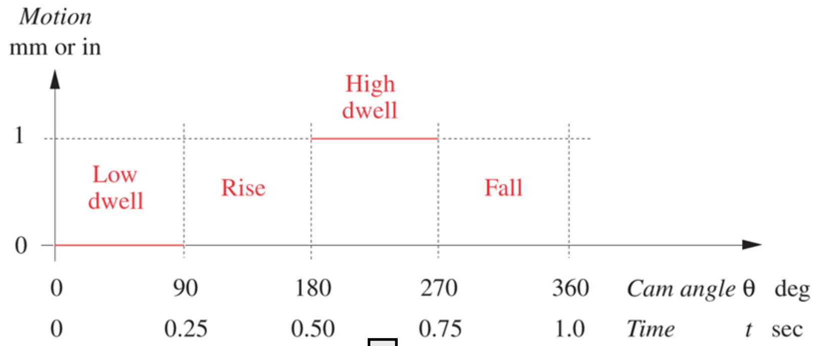
$$\theta = \omega\, t$$

Variables

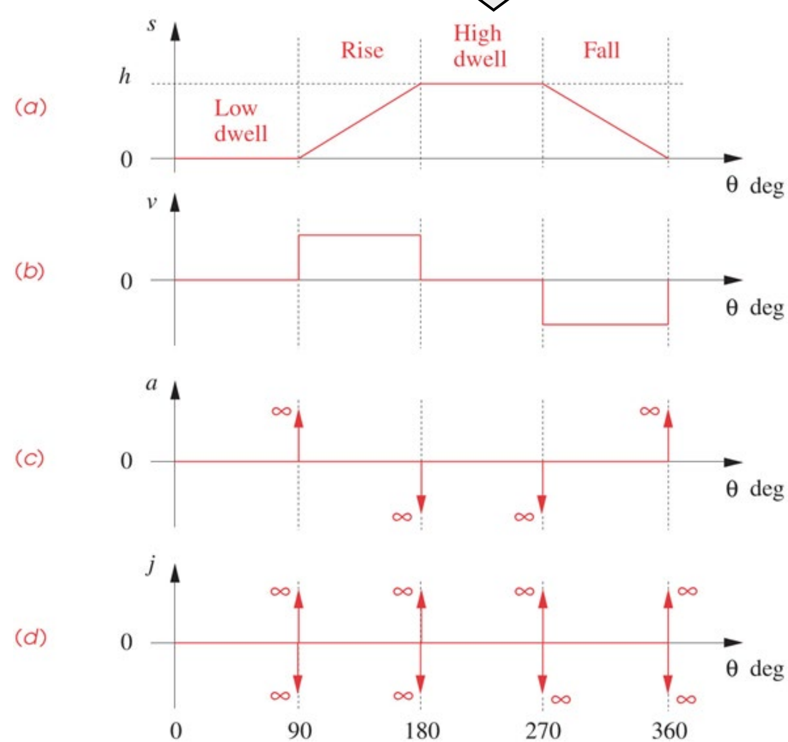
- t : time (sec)
- θ: cam angle (deg or rad)
- β: the angle over which the event occurs (deg or rad)
- θ/β: the normalized cam angle (unitless)
- ω: the camshaft angular velocity (rad/sec)
- *h:* lift/total displacement (length units)
- *s:* follower displacement (length units)
- *v:* follower velocity (length /rad)
- *a:* follower accel. (length/rad^2)
- *j:* follower jerk (length/rad^3)

Cam motion design –double dwell (DRDF)

- Cam timing diagram in a machine’s cycle (one revolution of the cam’s master driveshaft)



- SVAJ for a ***bad*** design



Discontinuities

???

- ## Law of cam design

The cam function must be ***continuous*** through the first and second derivatives of displacement , and the ***jerk function must be finite*** across the entire interval (360 degrees).

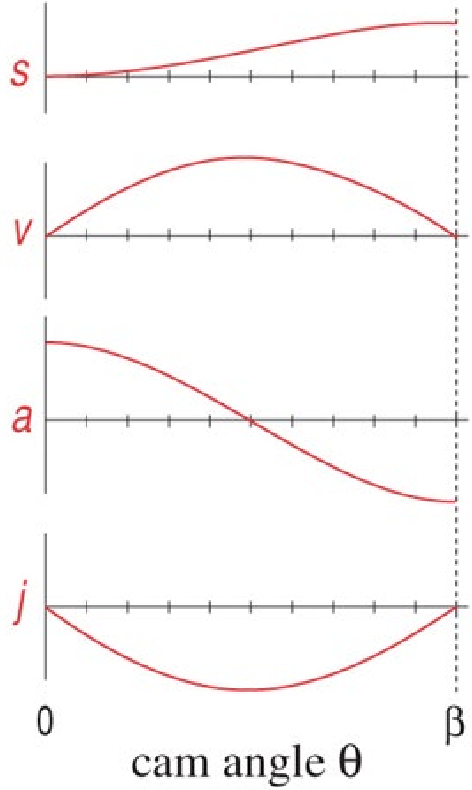
- ## Simple harmonic motion (SHM)

$$s = \frac{h}{2}\left[1 - \cos\left(\pi\frac{\theta}{\beta}\right)\right]$$

$$v = \frac{\pi}{\beta}\frac{h}{2}\sin\left(\pi\frac{\theta}{\beta}\right)$$

$$a = \frac{\pi^2}{\beta^2}\frac{h}{2}\cos\left(\pi\frac{\theta}{\beta}\right)$$

$$j = -\frac{\pi^3}{\beta^3}\frac{h}{2}\sin\left(\pi\frac{\theta}{\beta}\right)$$



h: rise
θ: cam shaft angle
β: total angle of the rise interval

For non-zero initial angle α,

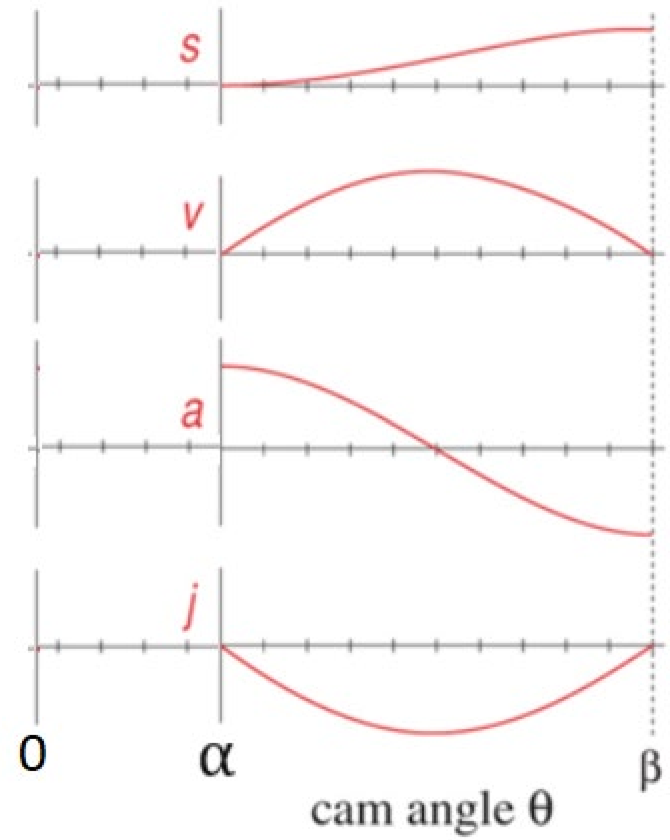
$$\theta' = \theta - \alpha$$

$$s = \frac{h}{2}\left[1 - \cos\left(\pi \frac{\theta'}{\beta}\right)\right]$$

$$v = \frac{\pi}{\beta}\frac{h}{2}\sin\left(\pi \frac{\theta'}{\beta}\right)$$

$$a = \frac{\pi^2}{\beta^2}\frac{h}{2}\cos\left(\pi \frac{\theta'}{\beta}\right)$$

$$j = -\frac{\pi^3}{\beta^3}\frac{h}{2}\sin\left(\pi \frac{\theta'}{\beta}\right)$$



A fall function can be derived by subtracting the rise displacement from the lift. For example,

$$s = h - \frac{h}{2}\left[1 - \cos\left(\pi \frac{\theta}{\beta}\right)\right]$$

*Note this only applies in **symmetric motion**.*

- **Example**: Sophomoric cam design –SHM

dwell	at zero displacement for 90 degrees (low dwell)
rise	1 in (25 mm) in 90 degrees
dwell	at 1 in (25 mm) for 90 degrees (high dwell)
fall	1 in (25 mm) in 90 degrees
cam ω	2π rad/sec = 1 rev/sec

For the rise segment, $\beta = \frac{\pi}{2}, h = 25\ (mm), \alpha = \frac{\pi}{2}$

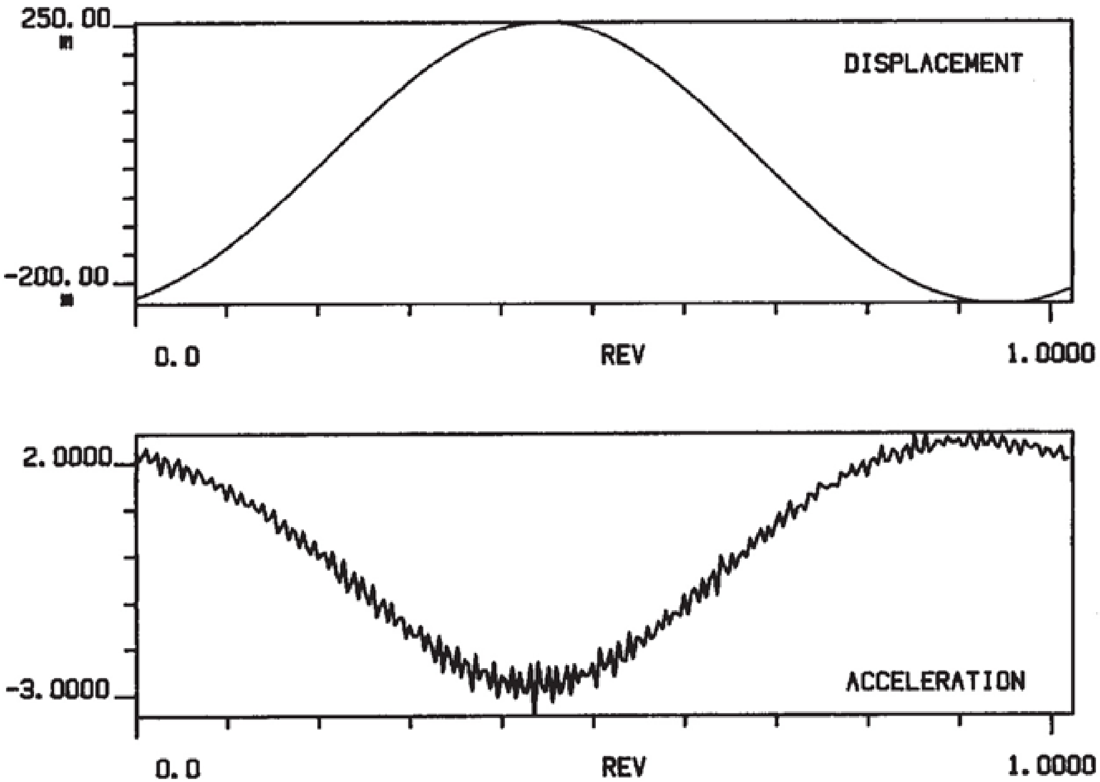
$$s = \frac{25}{2}\left[1 - \cos\left(\pi \frac{\theta - \frac{\pi}{2}}{\frac{\pi}{2}}\right)\right] = 12.5(1 - \cos(2\theta - \pi)), \frac{\pi}{2} \ll \theta < \pi$$

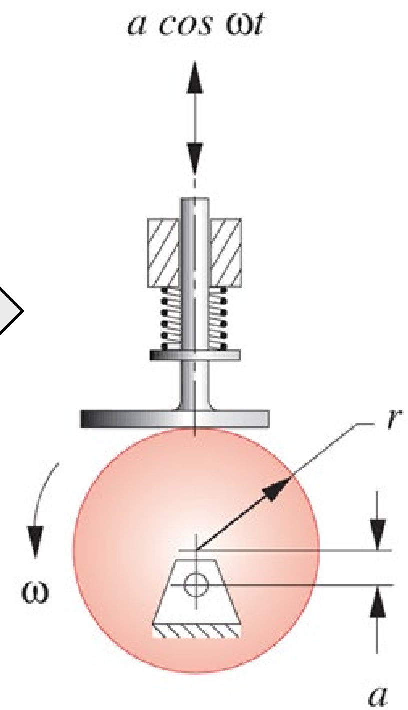
For the fall segment, $\beta = \frac{\pi}{2}, h = 25\ (mm), \alpha = \frac{3\pi}{2}$

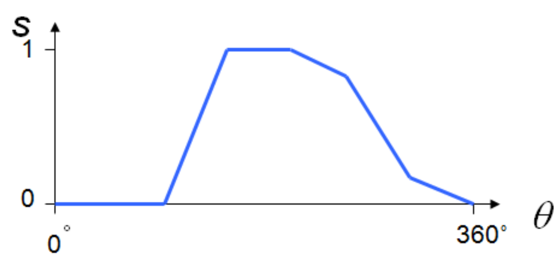
$$s = 25 - 12.5(1 - \cos(2\theta - 3\pi)),\ \frac{3\pi}{2} \ll \theta < 2\pi$$

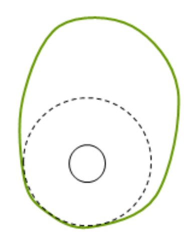
Non-zero acceleration at start and finish (***2.9m/sec^2***), but at the neighbouring dwell segments, it is zero; *these discontinuities are undesirable*. But, SHM is ok for RF motion (***without dwell***), meaning rise in 180 deg and then fall in 180 deg.

SHM motion implementation









Unacceptable Double-Dwell Functions

- pure constant velocity
- constant acceleration (parabolic displacement)
- cubic displacement
- simple harmonic motion

Some Acceptable Double-Dwell Functions

- Cycloidal
- Modified Trapezoidal
- Modified Sine

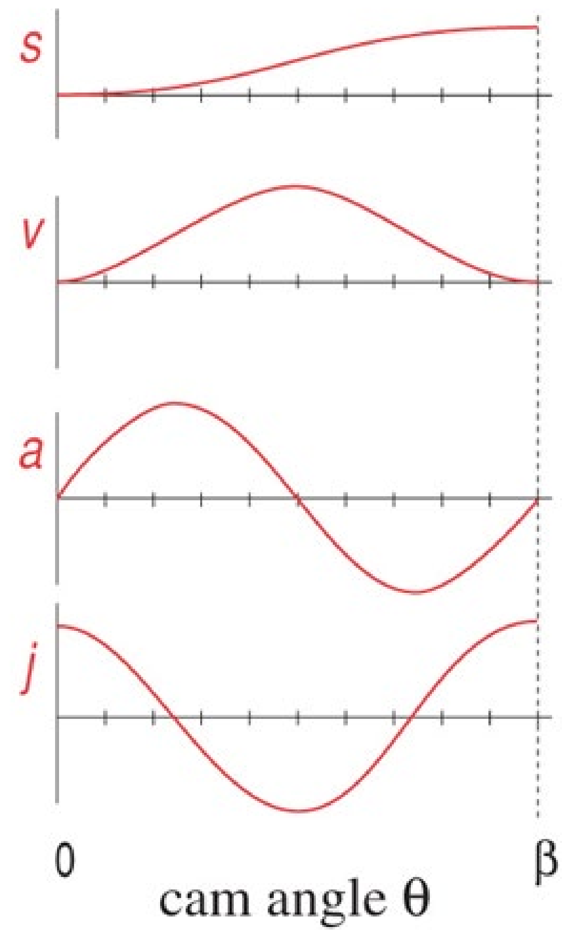
- **Cycloidal displacement**

$$s = h\left[\frac{\theta}{\beta} - \frac{1}{2\pi}\sin\left(2\pi\frac{\theta}{\beta}\right)\right]$$

$$v = \frac{h}{\beta}\left[1 - \cos\left(2\pi\frac{\theta}{\beta}\right)\right]$$

$$a = 2\pi\frac{h}{\beta^2}\sin\left(2\pi\frac{\theta}{\beta}\right)$$

$$j = 4\pi^2\frac{h}{\beta^3}\cos\left(2\pi\frac{\theta}{\beta}\right)$$



- **Example**: Junior cam design –cycloidal displacement

dwell	at zero displacement for 90 degrees (low dwell)
rise	1 in (25 mm) in 90 degrees
dwell	at 1 in (25 mm) for 90 degrees (high dwell)
fall	1 in (25 mm) in 90 degrees
cam ω	2π rad/sec = 1 rev/sec

For the rise segment, $\beta = \frac{\pi}{2}, h = 25\ (mm)$

$$s = 25\left(\frac{2\theta - \pi}{\pi} - \frac{1}{2\pi}\sin(4\theta - 2\pi)\right), \frac{\pi}{2} \ll \theta < \pi, \text{or}$$

$$s = 25\left(\frac{2\theta'}{\pi} - \frac{1}{2\pi}\sin(4\theta')\right),\ 0 \ll \theta' < \frac{\pi}{2},\ \theta' = \theta - \frac{\pi}{2}$$

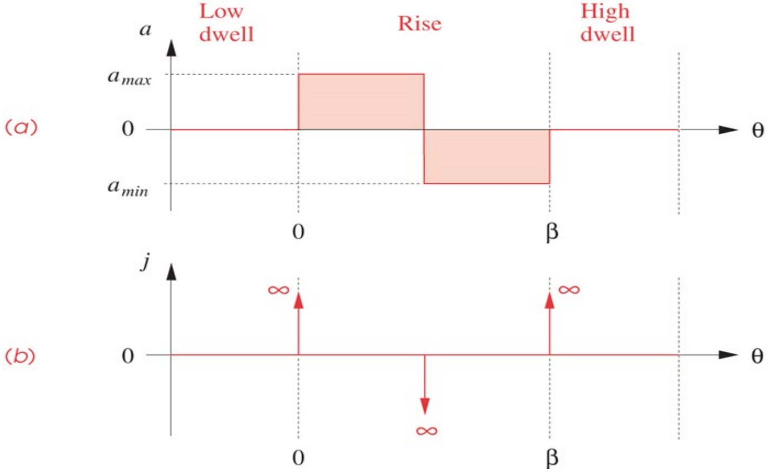
For the fall segment, $\beta = \frac{\pi}{2}, h = 25\ (mm)$

$$s = 25 - 25\left(\frac{2\theta'}{\pi} - \frac{1}{2\pi}\sin(4\theta')\right),\ 0 \ll \theta' < \frac{\pi}{2},\ \theta' = \theta - \frac{3\pi}{2}$$

Zero acceleration at start and finish; no discontinuities in velocity and acceleration. Peak acceleration (2.55 m/sec^2) and peak speed (0.2 m/sec) are little higher than those of other functions.

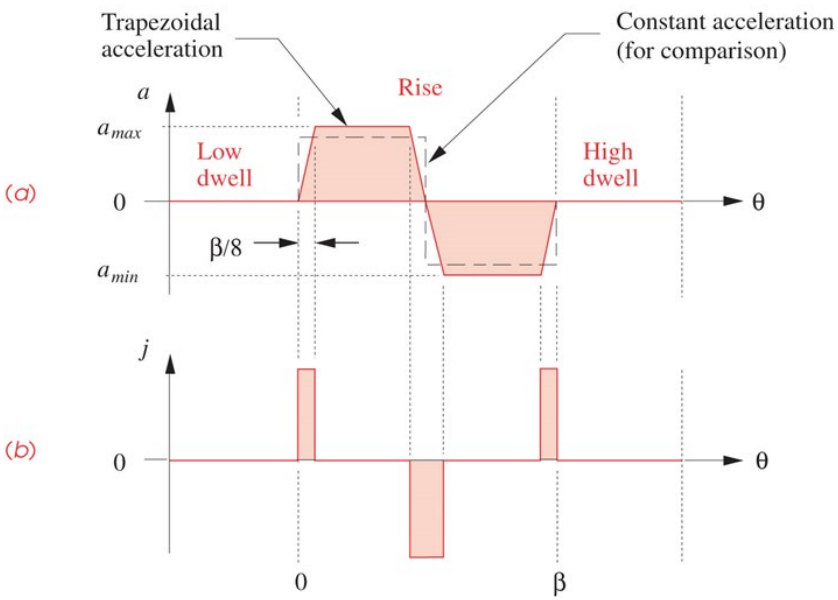
- **Combined functions**
  - **Constant acceleration**

    Discontinuity causes infinite jerk; *not acceptable*.

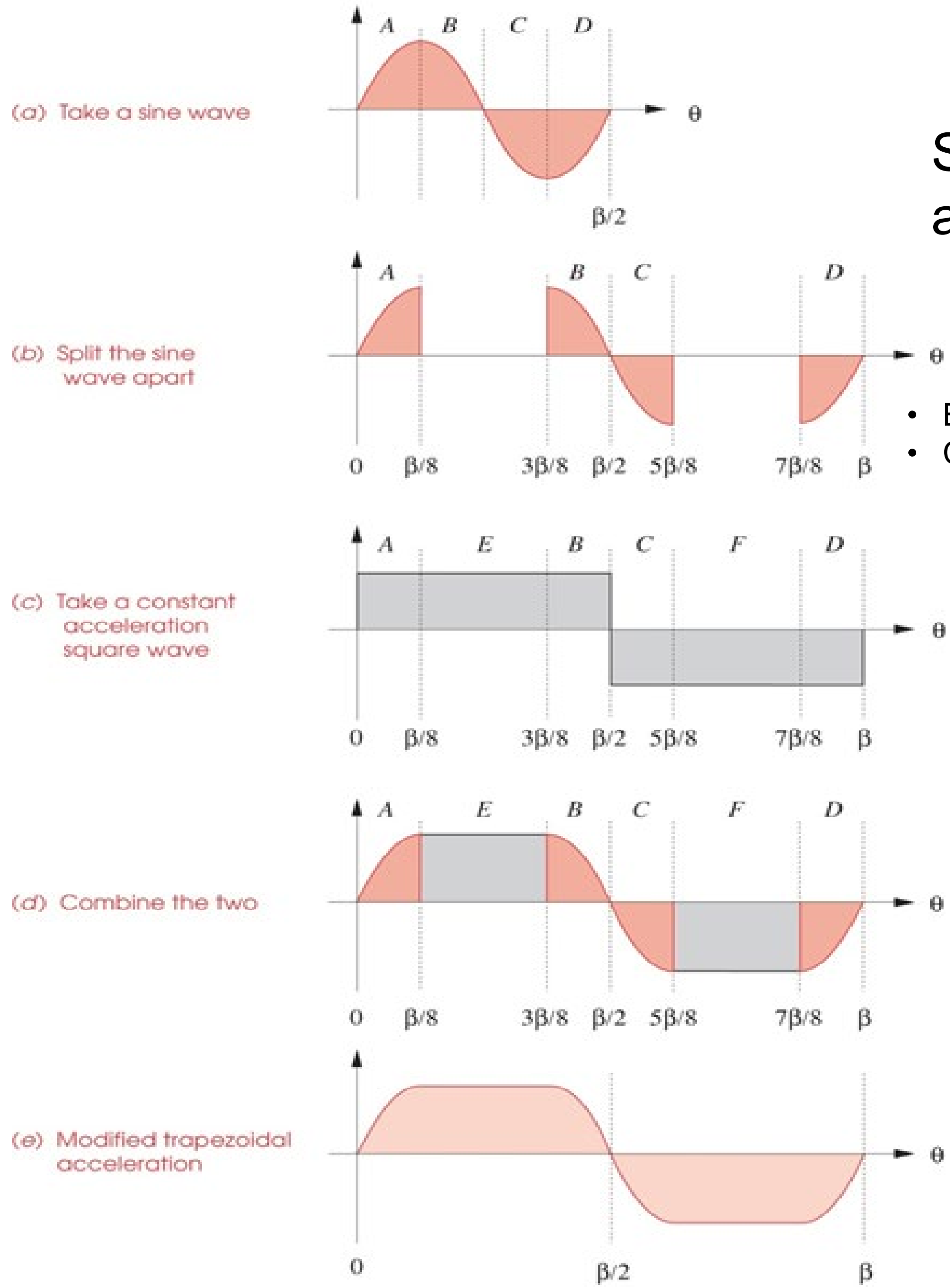


  - **Trapezoidal acceleration**

    Discontinuous in jerk; maximum acceleration can be less than that of SHM and cycloidal functions.



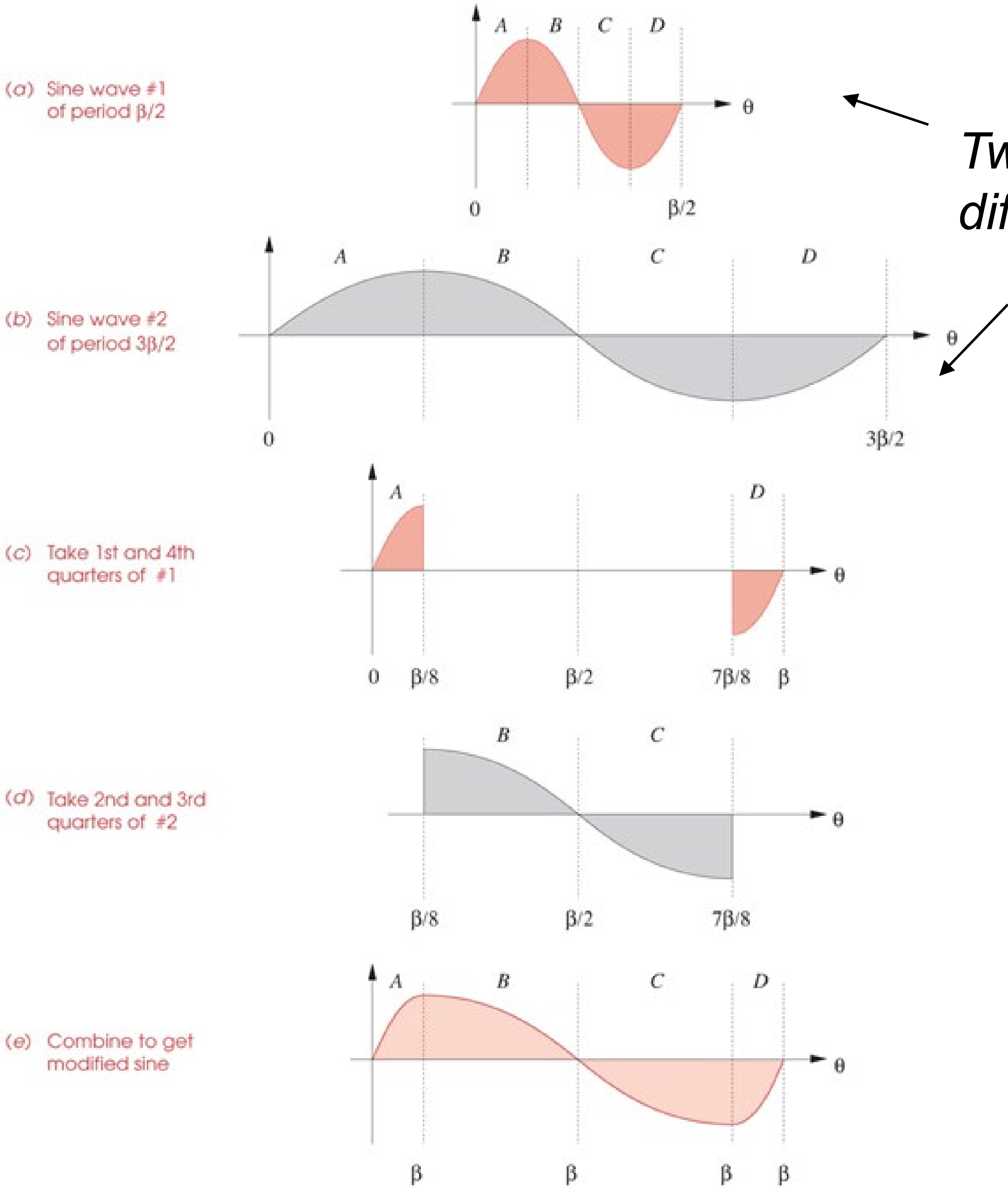
○ Modified trapezoidal acceleration



Sine and constant accelerations combined

- Each half interval is divided into 4 parts;
- Constant accl. for the two parts between

- **Modified sinusoidal acceleration (cycloidal displacement)**



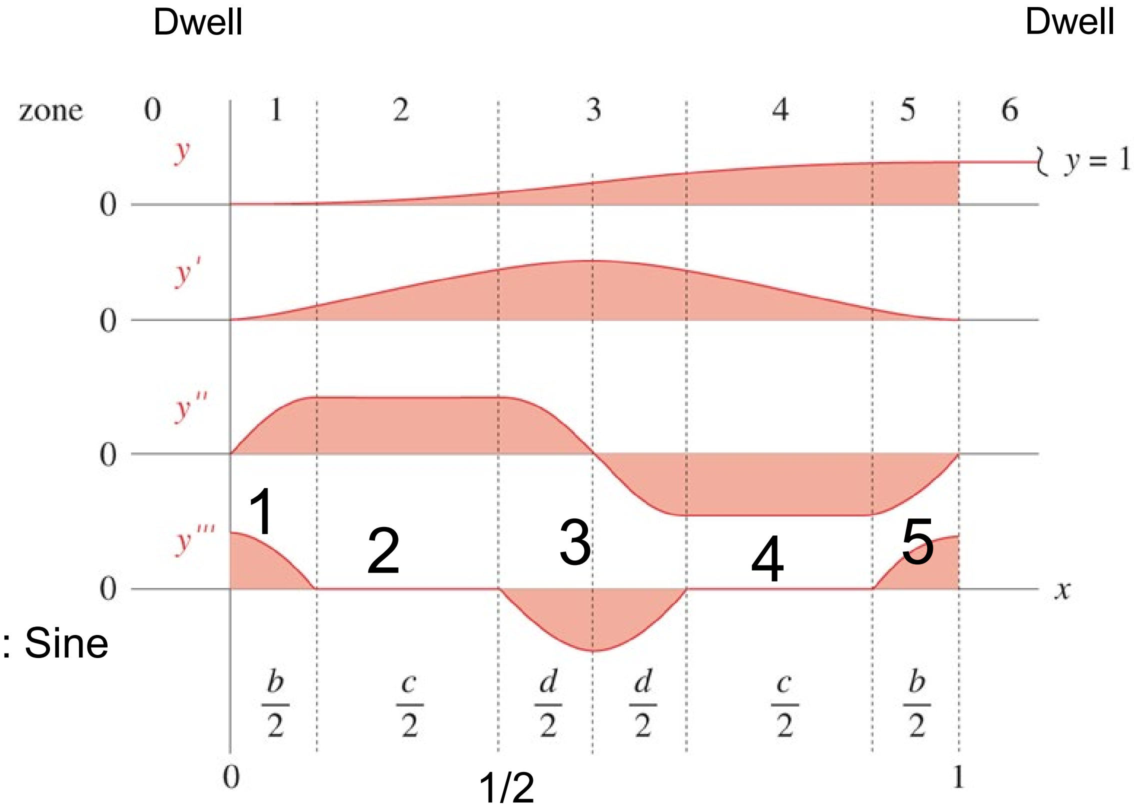
Two sinusoidal accelerations of different periods are combined.

○ The SCCA (Sine-Constant-Cosine-Acceleration) family of double–dwell functions

Normalization:

$$x = \frac{\theta}{\beta}, \quad y = \frac{s}{h}, \quad |x| \le 1,\ |y| \le 1$$

$$y' = \frac{dy}{dx}, \quad y'' = \frac{d^2 y}{dx^2}, \quad y''' = \frac{d^3 y}{dx^3}$$



$$b + c + d = 1$$

Zone 1: $0 \le x \le \frac{b}{2}$:

$$y = C_a\left[\frac{b}{\pi}x - \left(\frac{b}{\pi}\right)^2 \sin\left(\frac{\pi}{b}x\right)\right]$$, Acc: Sine

Zone 2: $\frac{b}{2} \le x \le \frac{1-d}{2}$

$$y = C_a\left[\frac{x^2}{2} + b\left(\frac{1}{\pi} - \frac{1}{2}\right)x + b^2\left(\frac{1}{8} - \frac{1}{\pi^2}\right)\right]$$, Acc: Constant

Acc: Cosine

Zone 3: $\frac{1-d}{2} \le x \le \frac{1+d}{2}$, $y = C_a\left\{\left(\frac{b}{\pi} + \frac{c}{2}\right)x + \left(\frac{d}{\pi}\right)^2 + b^2\left(\frac{1}{8} - \frac{1}{\pi^2}\right) - \frac{(1-d)^2}{8} - \left(\frac{d}{\pi}\right)^2 \cos\left[\frac{\pi}{d}\left(x - \frac{1-d}{2}\right)\right]\right\}$

Zone 4: $\dfrac{1+d}{2} \le x \le 1-\dfrac{b}{2}$, Acc: Costant

$$y = C_a\left[-\frac{x^2}{2}+\left(\frac{b}{\pi}+1-\frac{b}{2}\right)x+\left(2d^2-b^2\right)\left(\frac{1}{\pi^2}-\frac{1}{8}\right)-\frac{1}{4}\right]$$

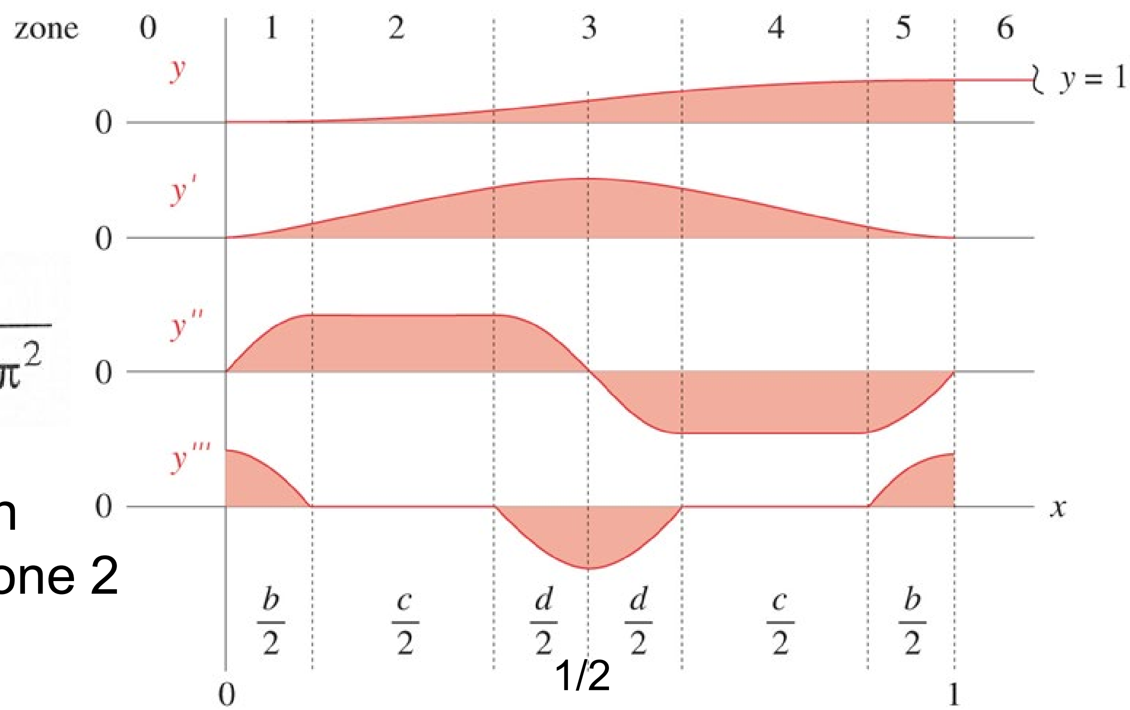
Zone 5: $1-\dfrac{b}{2} \le x \le 1$, Acc: Sine

$$y = C_a\left\{\frac{b}{\pi}x+\frac{2\left(d^2-b^2\right)}{\pi^2}+\frac{(1-b)^2-d^2}{4}-\left(\frac{b}{\pi}\right)^2\sin\left[\frac{\pi}{b}(x-1)\right]\right\}$$

At the end of Zone 5, x=1, displacement, y= 1

$$C_a = \frac{4\pi^2}{\left(\pi^2-8\right)\left(b^2-d^2\right)-2\pi(\pi-2)b+\pi^2}$$

This is also the peak acceleration at the zones of constant acc.-Zone 2 and Zone 4.

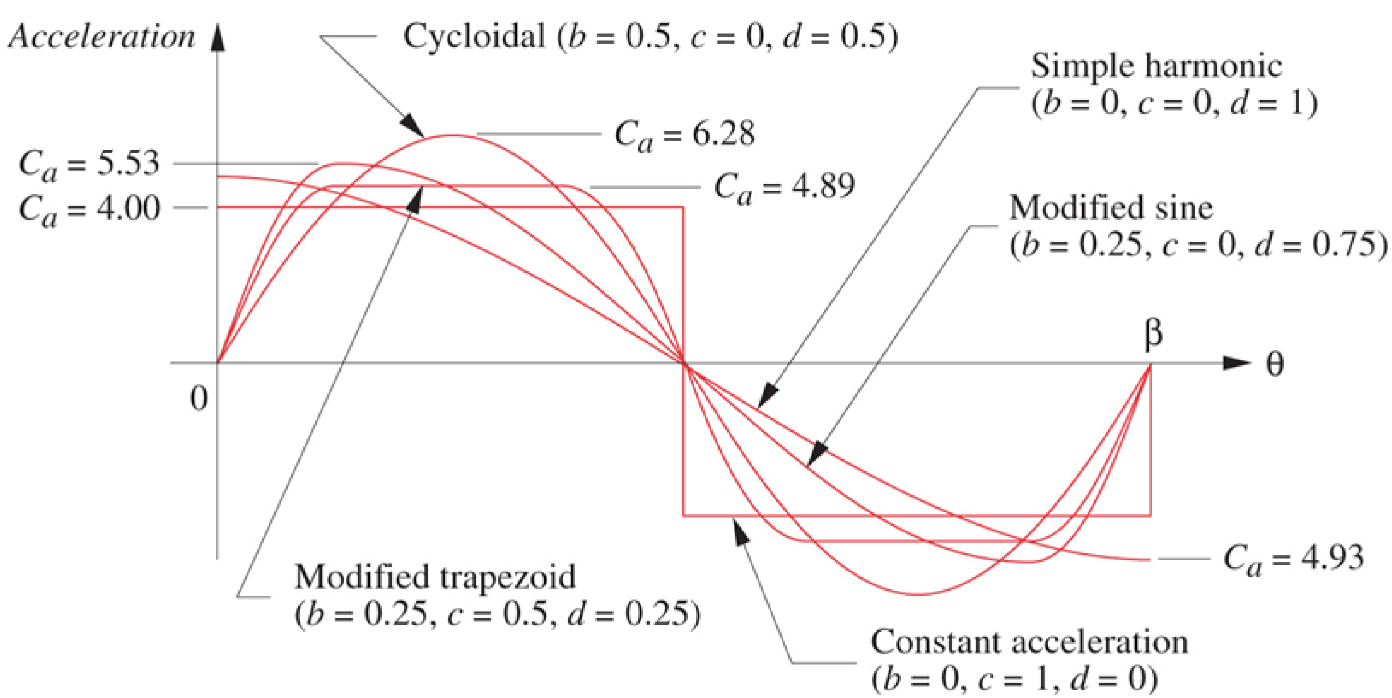


Peak velocity at x=0.5: $C_v = C_a\left(\frac{b+d}{\pi} + \frac{c}{2}\right)$

Peak jerk at x=0: $C_j = C_a \frac{\pi}{b} \qquad b \neq 0$

Parameters and Coefficients for the SCCA Family of Functions

Function	b	c	d	C_v	C_a	C_j
Constant acceleration	0.00	1.00	0.00	2.0000	4.0000	infinite
Modified trapezoid	0.25	0.50	0.25	2.0000	4.8881	61.426
Simple harmonic	0.00	0.00	1.00	1.5708	4.9348	infinite
Modified sine	0.25	0.00	0.75	1.7596	5.5280	69.466
Cycloidal displacement	0.50	0.00	0.50	2.0000	6.2832	39.478



SVAJ

$$s = hy \qquad v = \frac{h}{\beta} y' , \qquad a = \frac{h}{\beta^2} y'' \qquad j = \frac{h}{\beta^3} y'''$$

- **Example**: Senior cam design –combined function

dwell	at zero displacement for 90 degrees (low dwell)
rise	1 in (25 mm) in 90 degrees
dwell	at 1 in (25 mm) for 90 degrees (high dwell)
fall	1 in (25 mm) in 90 degrees
cam ω	2π rad/sec = 1 rev/sec

It can be checked that both modified trapezoidal or modified sinusoid functions are acceptable. For the modified trapezoidal function, the peak acceleration is 1.98m/s^2, peak jerk is 100m/s^3; for the modified sinusoidal function, the peak acceleration is 2.24 m/s^2 and the peak jerk is 113m/s^3.

- **Polynomial functions - more versatile functions**

$$s = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + C_5 x^5 + C_6 x^6 + \cdots + C_n x^n$$

where

s: follower's displacement

x: independent variable which can be θ/β or time

C_i: unknown coefficients to be determined

The key question here is the boundary conditions (BC) to be specified on SVAJ.

The number of BCs determine the order of the polynomial.

○ **Example**: 3-4-5 polynomial for double-dwell case

dwell	at zero displacement for 90 degrees (low dwell)
rise	1 in (25 mm) in 90 degrees
dwell	at 1 in (25 mm) for 90 degrees (high dwell)
fall	1 in (25 mm) in 90 degrees
cam ω	2π rad/sec = 1 rev/sec

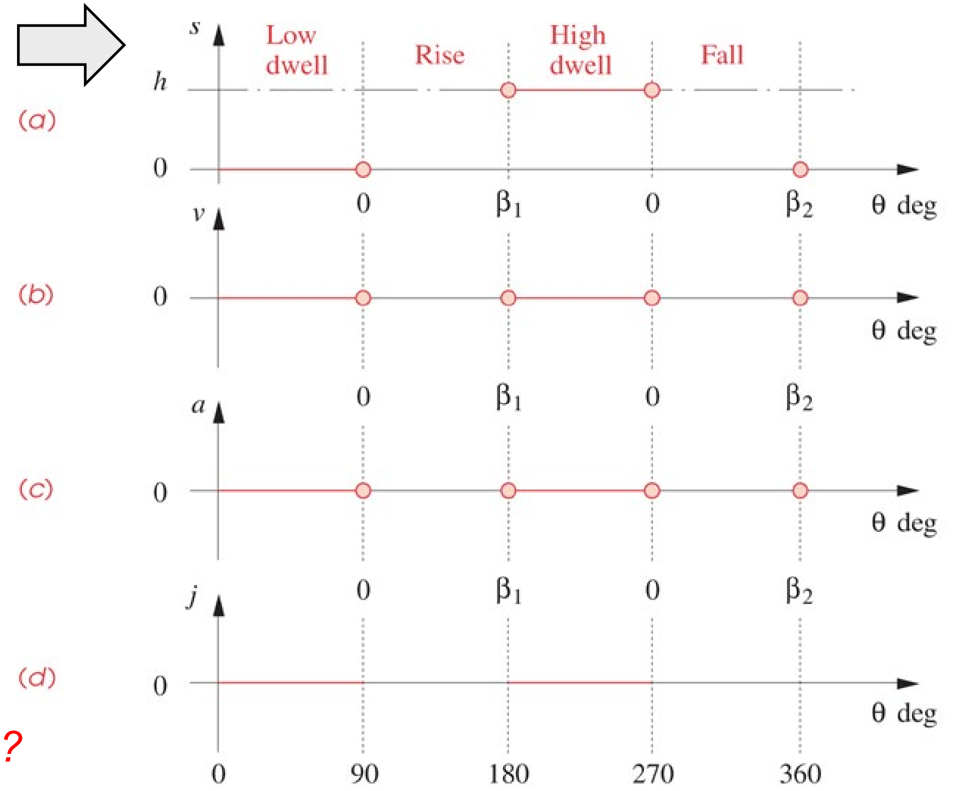
There are 6 BCs each for the rise and the fall segments , and then a 5th order polynomial is needed:

$$s = C_0 + C_1\left(\frac{\theta}{\beta}\right) + C_2\left(\frac{\theta}{\beta}\right)^2 + C_3\left(\frac{\theta}{\beta}\right)^3 + C_4\left(\frac{\theta}{\beta}\right)^4 + C_5\left(\frac{\theta}{\beta}\right)^5$$

$$v = \frac{1}{\beta}\left[C_1 + 2C_2\left(\frac{\theta}{\beta}\right) + 3C_3\left(\frac{\theta}{\beta}\right)^2 + 4C_4\left(\frac{\theta}{\beta}\right)^3 + 5C_5\left(\frac{\theta}{\beta}\right)^4\right]$$

$$a = \frac{1}{\beta^2}\left[2C_2 + 6C_3\left(\frac{\theta}{\beta}\right) + 12C_4\left(\frac{\theta}{\beta}\right)^2 + 20C_5\left(\frac{\theta}{\beta}\right)^3\right]$$

How many BCs are there if jerk is considered ?



Boundary conditions

From BCs:

$$\theta = 0,\ s = 0 \Rightarrow C_0 = 0$$

$$\theta = 0,\ v = 0 \Rightarrow C_1 = 0$$

$$\theta = 0,\ a = 0 \Rightarrow C_2 = 0$$

$$\theta = \beta,\ s = h \Rightarrow h = C_3 + C_4 + C_5$$

$$\theta = \beta,\ v = 0 \Rightarrow 0 = \frac{1}{\beta}\left[3C_3 + 4C_4 + 5C_5\right]$$

$$\theta = \beta,\ a = 0 \Rightarrow 0 = \frac{1}{\beta^2}\left[6C_3 + 12C_4 + 20C_5\right]$$

$$\Rightarrow C_3 = 10h, \quad C_4 = -15h, \quad C_5 = 6h$$

$$s = h\left[10\left(\frac{\theta}{\beta}\right)^3 - 15\left(\frac{\theta}{\beta}\right)^4 + 6\left(\frac{\theta}{\beta}\right)^5\right]$$

3-4-5 Polynomial

$$s = h\left[10\left(\frac{\theta}{\beta}\right)^3 - 15\left(\frac{\theta}{\beta}\right)^4 + 6\left(\frac{\theta}{\beta}\right)^5\right]$$

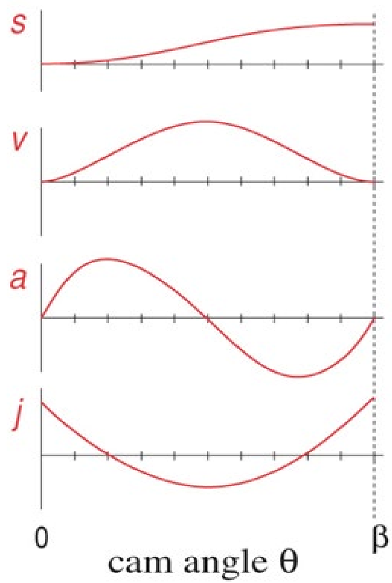
3-4-5 Polynomial

$$s = h\left[35\left(\frac{\theta}{\beta}\right)^4 - 84\left(\frac{\theta}{\beta}\right)^5 + 70\left(\frac{\theta}{\beta}\right)^6 - 20\left(\frac{\theta}{\beta}\right)^7\right]$$

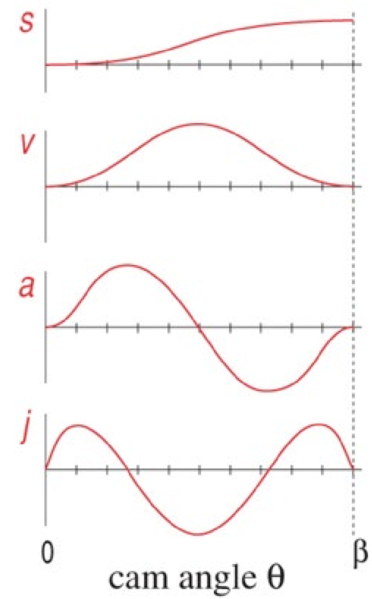
4-5-6-7 Polynomial

(jerk constraints added)

(**8 conditions** !)

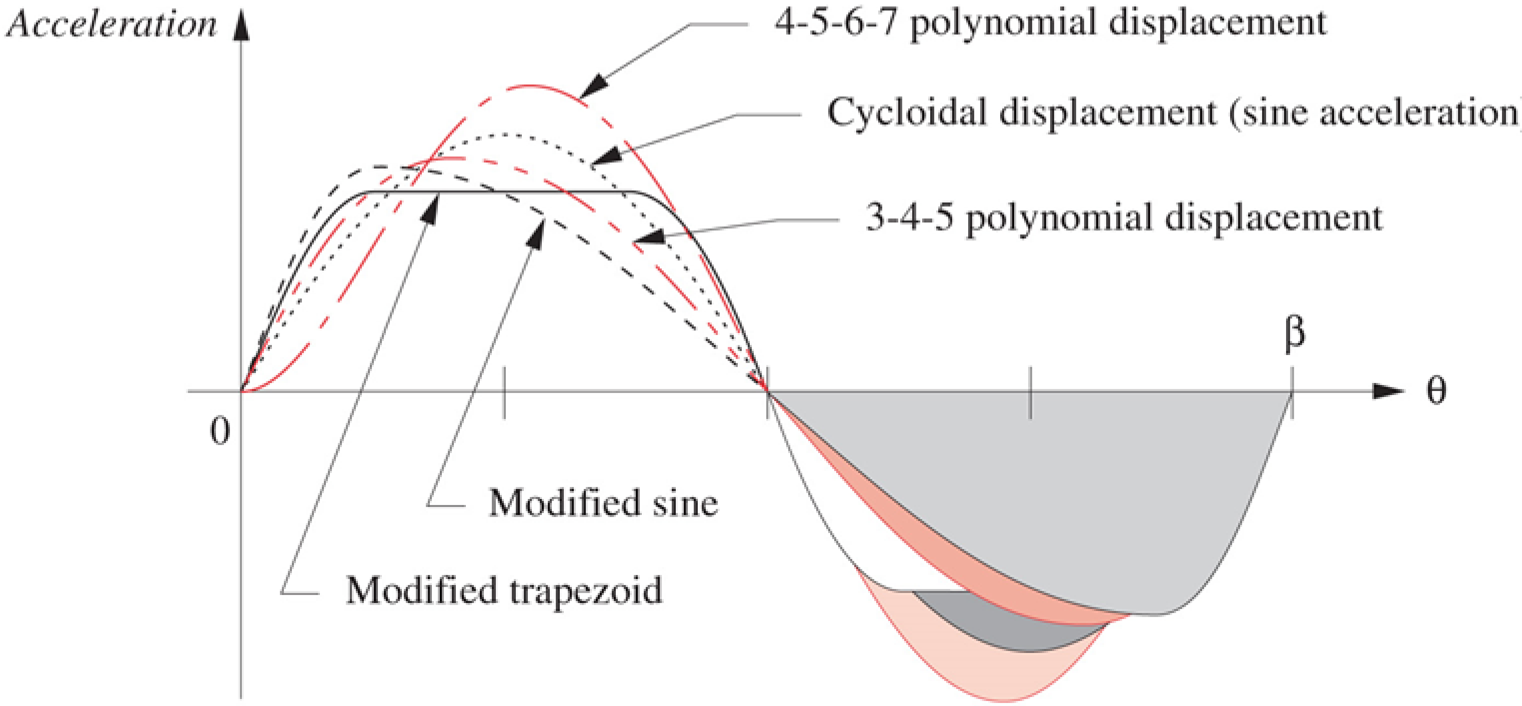


3-4-5 Polynomial

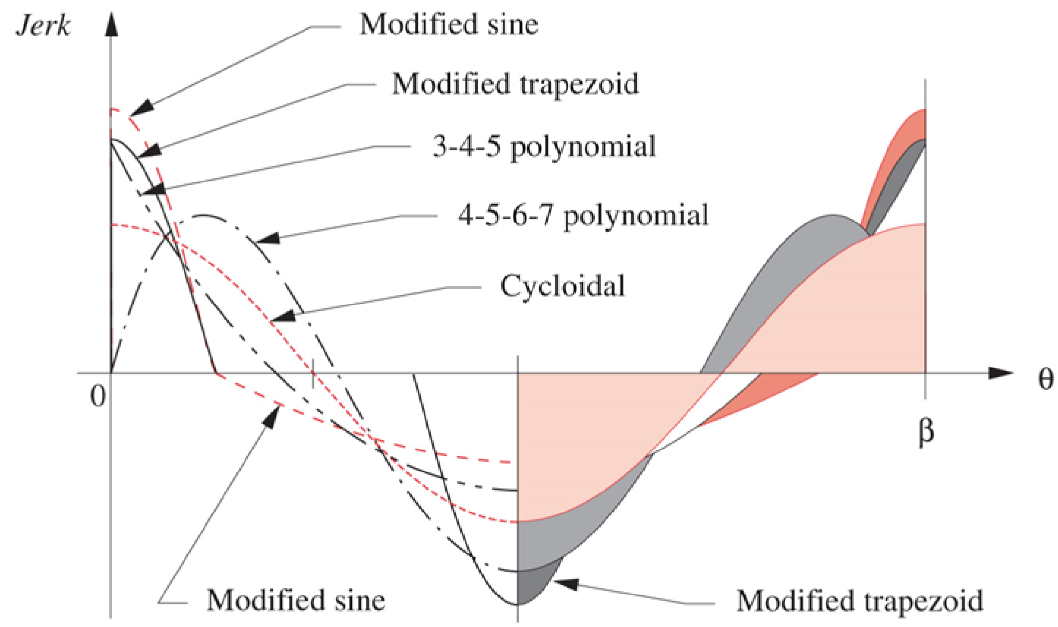


4-5-6-7 Polynomial

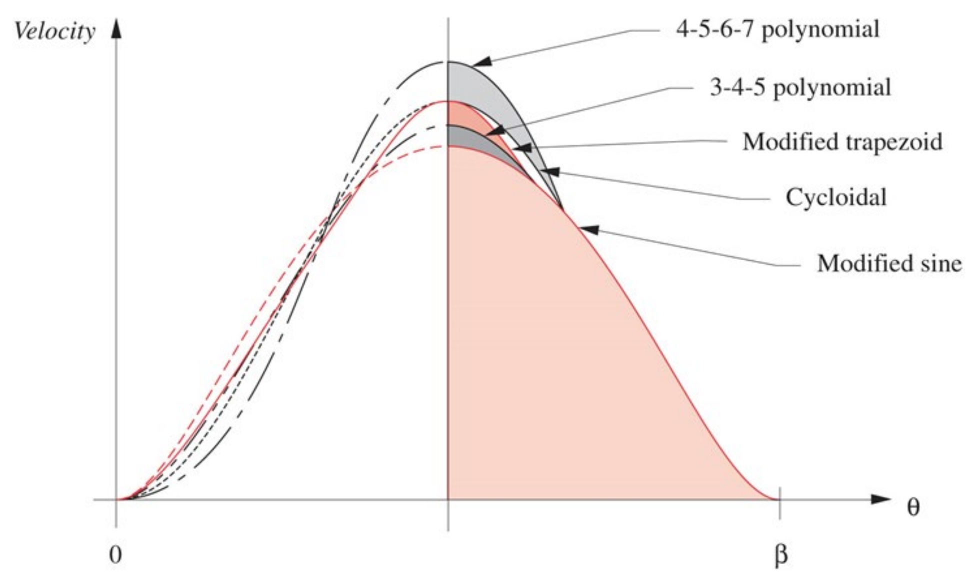
Comparison of accelerations from various functions:



Comparison of jerks :



Comparison of velocities:

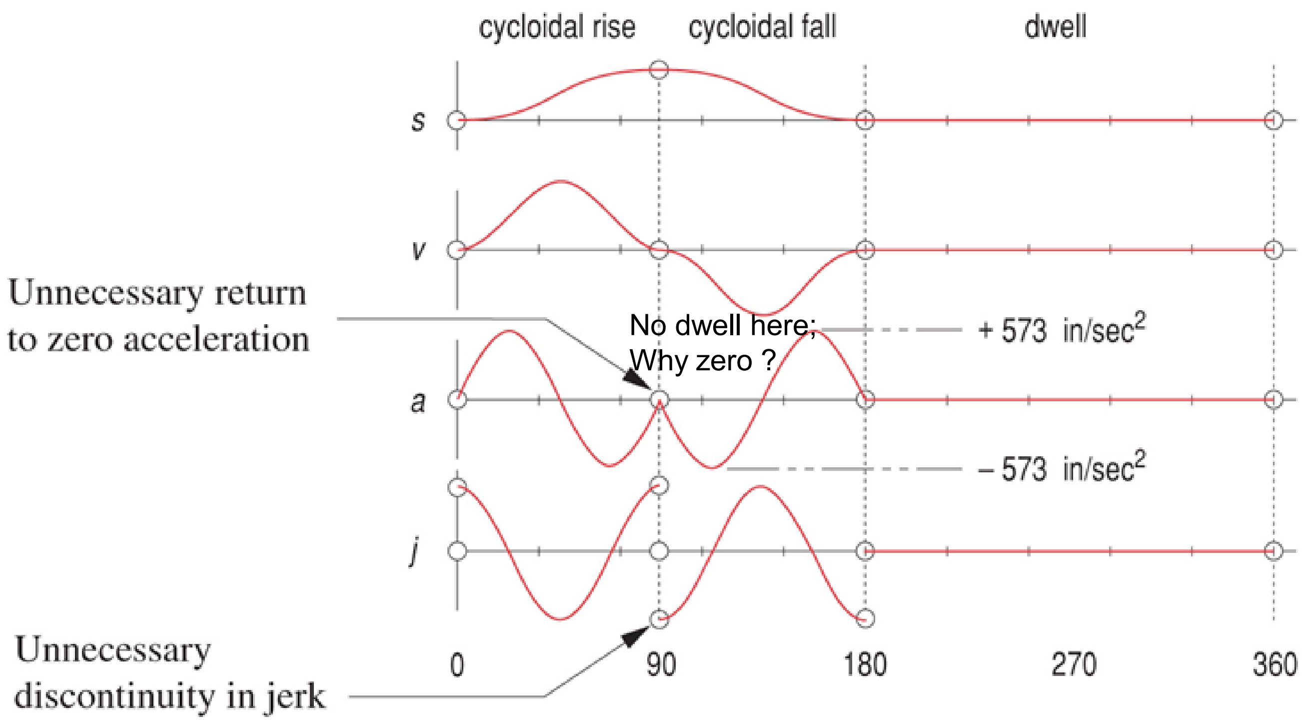


Factors for Various Functions

Function	Max. Veloc.	Max. Accel.	Max. Jerk	Comments
Constant accel.	2.000 h/β	4.000 h/β^2	Infinite	∞ jerk—not acceptable
Harmonic disp.	1.571 h/β	4.945 h/β^2	Infinite	∞ jerk—not acceptable
Trapezoid accel.	2.000 h/β	5.300 h/β^2	44 h/β^3	Not as good as mod. trap
Mod. trap. accel.	2.000 h/β	4.888 h/β^2	61 h/β^3	Low accel. but rough jerk
Mod. sine accel.	1.760 h/β	5.528 h/β^2	69 h/β^3	Low veloc., good accel
3-4-5 poly. disp.	1.875 h/β	5.777 h/β^2	60 h/β^3	Good compromise
Cycloidal disp.	2.000 h/β	6.283 h/β^2	40 h/β^3	Smooth accel. and jerk.
4-5-6-7 poly. disp.	2.188 h/β	7.526 h/β^2	52 h/β^3	Smooth jerk, high accel.

Cam motion design – single dwell (RFD)

- Cycloidal motion -- a poor choice:



Rule of thumb: the acceleration does not return to zero at the end of rise interval, maintain non-zero at the beginning and becomes zero at the end of fall interval.

- Double harmonic – full period for one harmonic and half period for another:

Rise :

$$s=\frac{h}{2}\left\{\left[1-\cos\left(\pi\frac{\theta}{\beta}\right)\right]-\frac{1}{4}\left[1-\cos\left(2\pi\frac{\theta}{\beta}\right)\right]\right\}$$

$$v=\frac{\pi}{\beta}\frac{h}{2}\left[\sin\left(\pi\frac{\theta}{\beta}\right)-\frac{1}{2}\sin\left(2\pi\frac{\theta}{\beta}\right)\right]$$

$$a=\frac{\pi^2}{\beta^2}\frac{h}{2}\left[\cos\left(\pi\frac{\theta}{\beta}\right)-\cos\left(2\pi\frac{\theta}{\beta}\right)\right]$$

$$j=-\frac{\pi^3}{\beta^3}\frac{h}{2}\left[\sin\left(\pi\frac{\theta}{\beta}\right)-2\sin\left(2\pi\frac{\theta}{\beta}\right)\right]$$

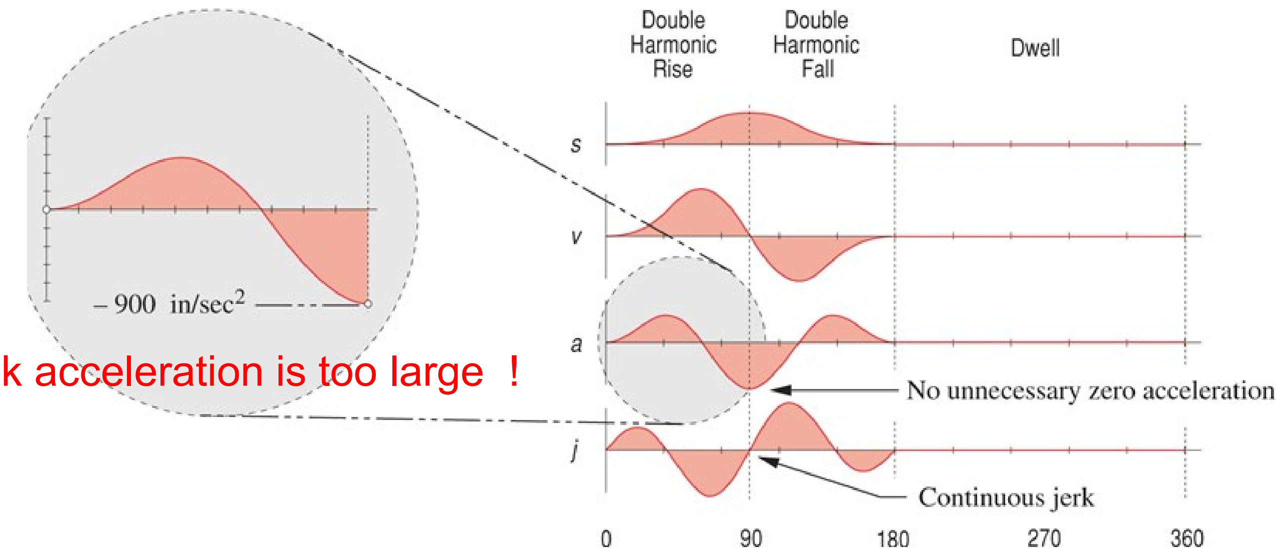
Fall:

$$s=\frac{h}{2}\left\{\left[1+\cos\left(\pi\frac{\theta}{\beta}\right)\right]+\frac{1}{4}\left[1-\cos\left(2\pi\frac{\theta}{\beta}\right)\right]\right\}$$

$$v=-\frac{\pi}{\beta}\frac{h}{2}\left[\sin\left(\pi\frac{\theta}{\beta}\right)+\frac{1}{2}\sin\left(2\pi\frac{\theta}{\beta}\right)\right]$$

$$a=-\frac{\pi^2}{\beta^2}\frac{h}{2}\left[\cos\left(\pi\frac{\theta}{\beta}\right)+\cos\left(2\pi\frac{\theta}{\beta}\right)\right]$$

$$j=\frac{\pi^3}{\beta^3}\frac{h}{2}\left[\sin\left(\pi\frac{\theta}{\beta}\right)+2\sin\left(2\pi\frac{\theta}{\beta}\right)\right]$$



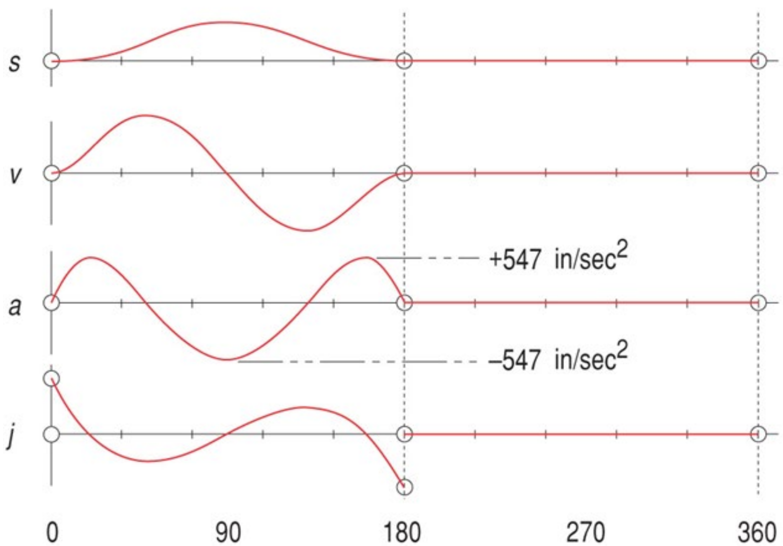
Peak acceleration is too large !

- Polynomials– one function for rise and fall (minimization of the number of segments and the boundary conditions (BCs))

Symmetrical rise-fall:

rise-fall	1 in (25 mm) in 90° and fall 1 in (25 mm) in 90° for a total of 180°
dwell	at zero displacement for 180° (low dwell)

$$s = h\left[64\left(\frac{\theta}{\beta}\right)^3 - 192\left(\frac{\theta}{\beta}\right)^4 + 192\left(\frac{\theta}{\beta}\right)^5 - 64\left(\frac{\theta}{\beta}\right)^6\right]$$



β

Segment number	Function used	Start angle	End angle	Delta angle
1	Poly 6	0	180	180

Boundary Conditions Imposed

Function	Theta	% Beta	Boundary Cond.
Displ	0	0	0
Veloc	0	0	0
Accel	0	0	0
Displ	180	1	0
Veloc	180	1	0
Accel	180	1	0
Displ	90	0.5	1

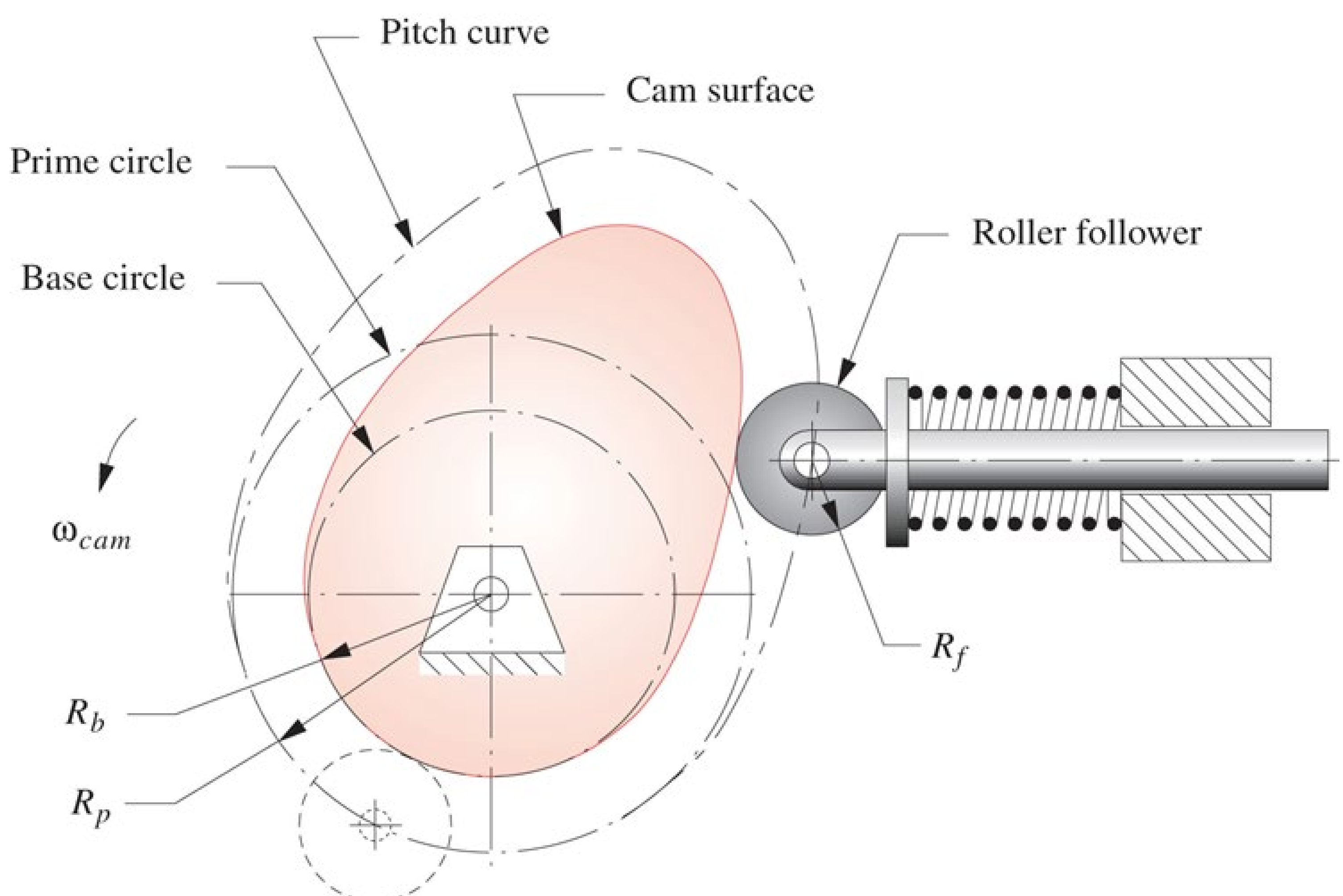
Equation Resulting

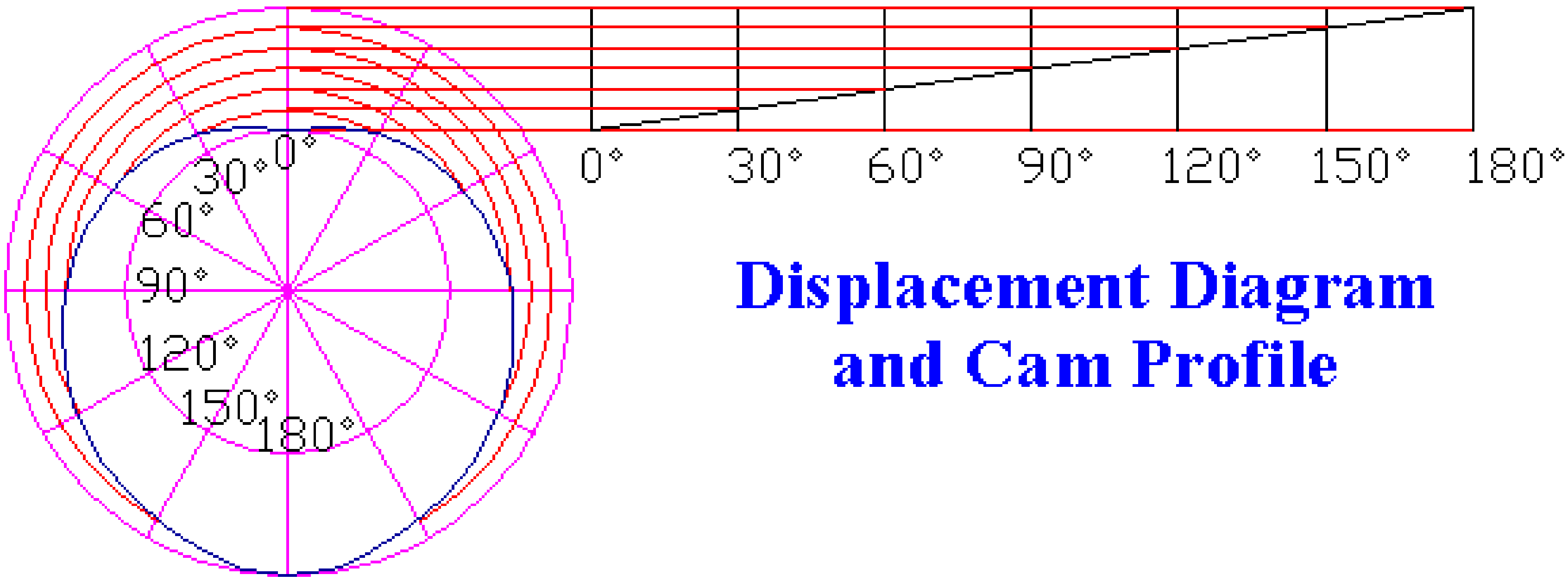
Exponent	Coefficient
0	0
1	0
2	0
3	64
4	– 192
5	192
6	– 64

Boundary conditions

- S=V= A =0 at each end of the rise fall segment
- Peak rise at 90 deg.

SVAJ → Sizing the cam – pressure angle and radius of curvature





Displacement Diagram and Cam Profile

TABLE 8.6 Follower Displacements for Example 8.3

θ	0, 360°	10°	20°	30°	40°	50°	60°	70°	80°
y	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
θ	90°	100°	110°	120°	130°	140°	150°	160°	170°
y	0.0000	0.0000	0.0000	0.0000	0.0536	0.2000	0.4000	0.6000	0.7464
θ	180°	190°	200°	210°	220°	230°	240°	250°	260°
y	0.8000	0.8000	0.8000	0.8000	0.7913	0.7654	0.7236	0.6677	0.6000
θ	270°	280°	290°	300°	310°	320°	330°	340°	350°
y	0.5236	0.4418	0.3582	0.2764	0.2000	0.1323	0.0764	0.0346	0.0087

Follower displacement

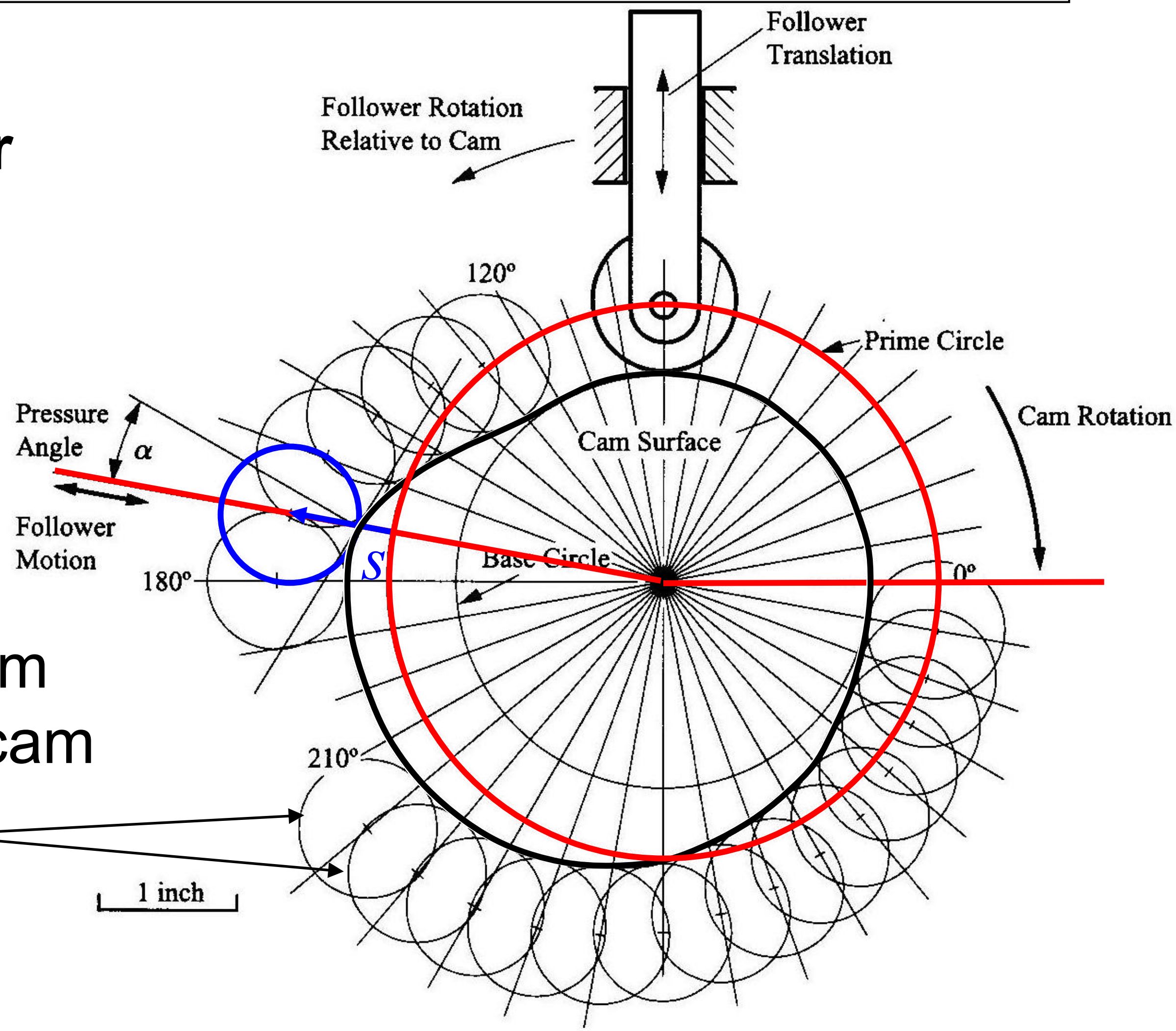
$$y = s(\theta)$$

Graphic design – radial roller follower

$\theta = 170^\circ$

$s = 0.7464$

the roller positions form an 'envelope' for the cam profile



Key terms

- *Base circle*: the smallest circle tangent to the physical surface of the cam
- *Prime circle*: the smallest circle tangent to the locus of the centreline of the follower
- *Pressure angle*: the angle between the follower velocity and the normal of the cam surface at the contact point, or *axis of transmission* which is perpendicular to the *axis of slip* (common tangent). It is usually up to 30 ~35 deg.

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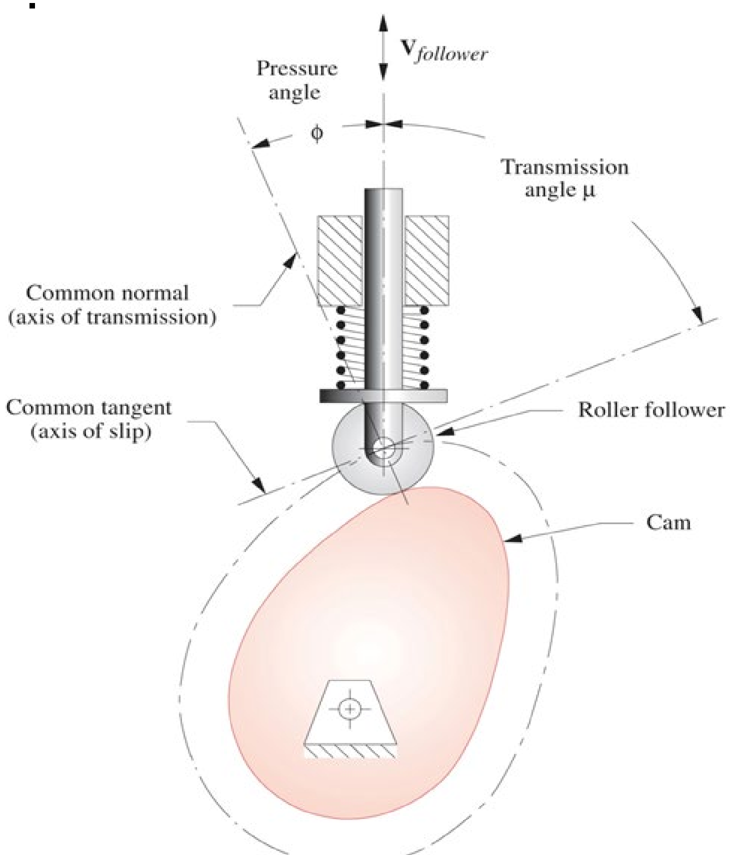
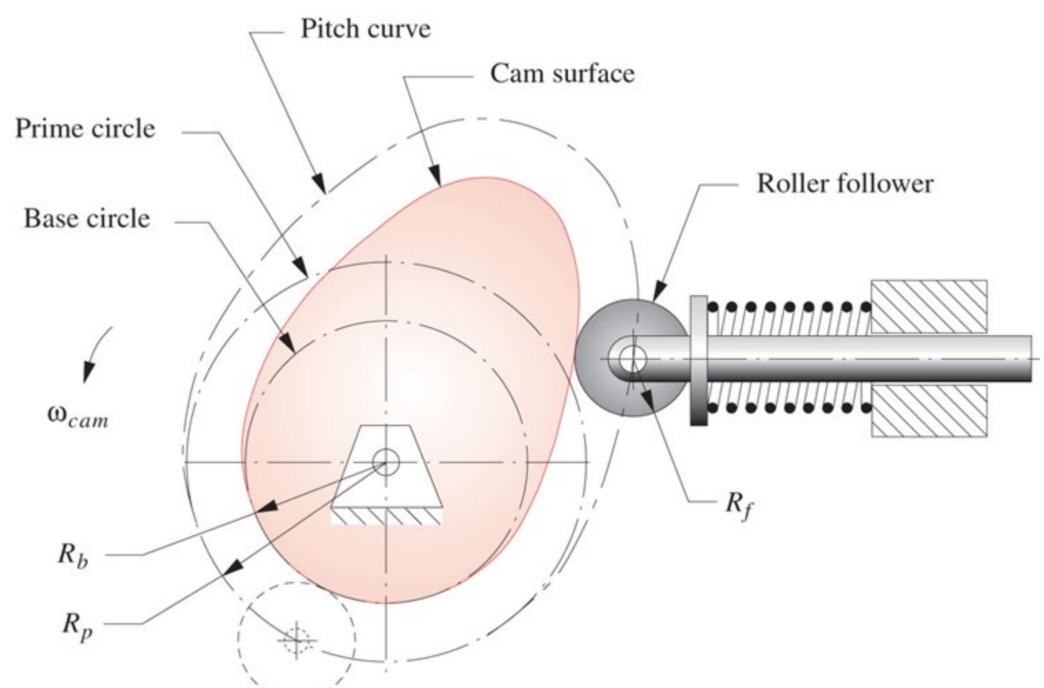




- *Eccentricity* (ε): the distance between the follower's axis and the centre of the cam. When it is zero, the follower is called *aligned follower*

$$V_{I_{2,4}} = b\omega = \frac{ds}{dt} = \frac{ds}{d\theta}\frac{d\theta}{dt} = \frac{ds}{d\theta}\omega = v\omega$$

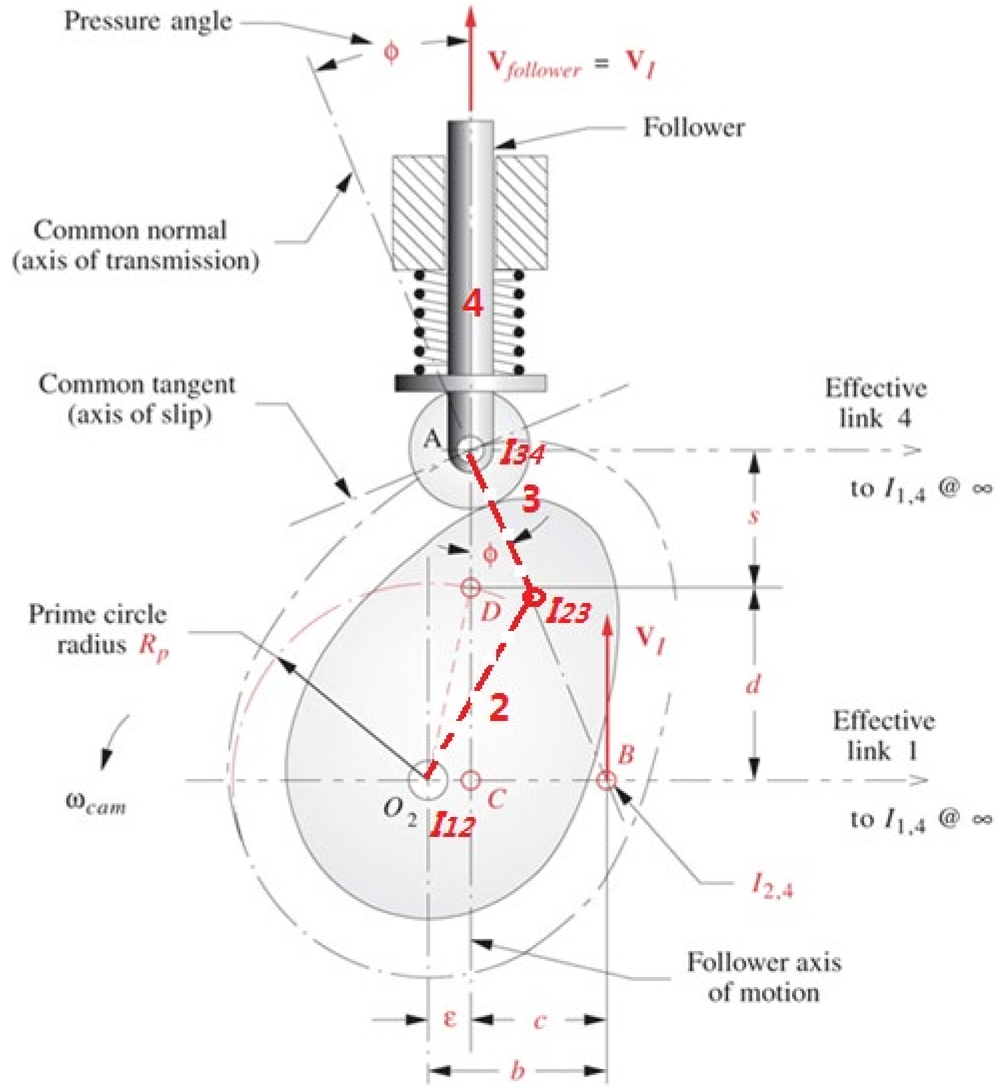
$$b\omega = v\omega \Rightarrow b = v$$

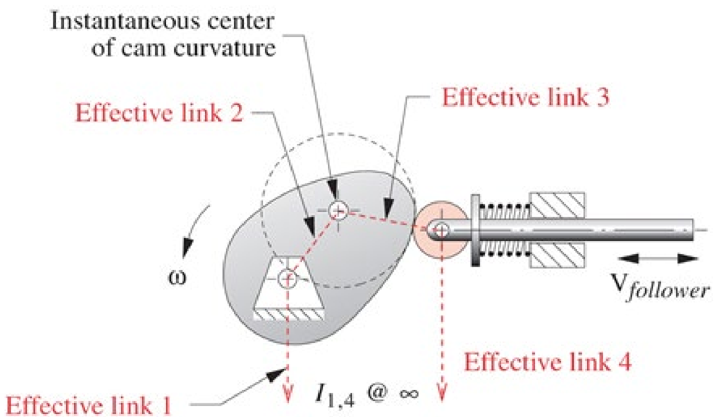
$$c = (s+d)\tan\phi \qquad \text{(Triangle ABC)}$$

$$b = (s+d)\tan\phi + \varepsilon \Rightarrow v = (s+d)\tan\phi + \varepsilon$$

$$d = \sqrt{R_P^2 - \varepsilon^2}$$

$$\phi = \arctan\frac{v-\varepsilon}{s+\sqrt{R_P^2-\varepsilon^2}}$$



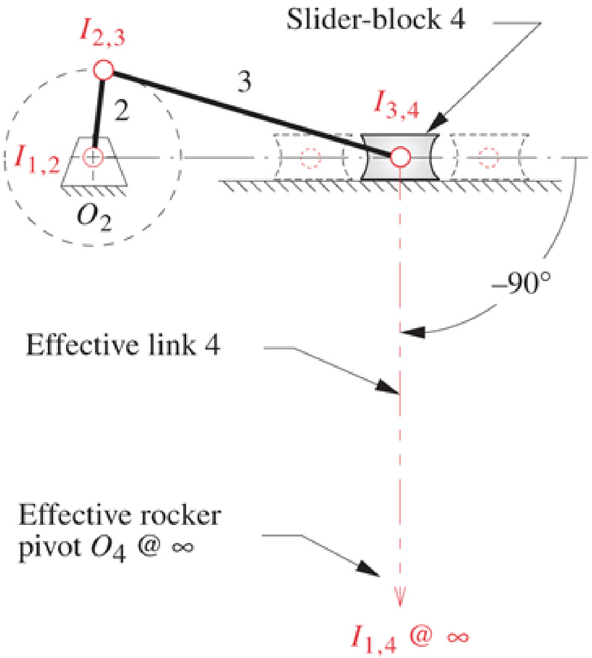


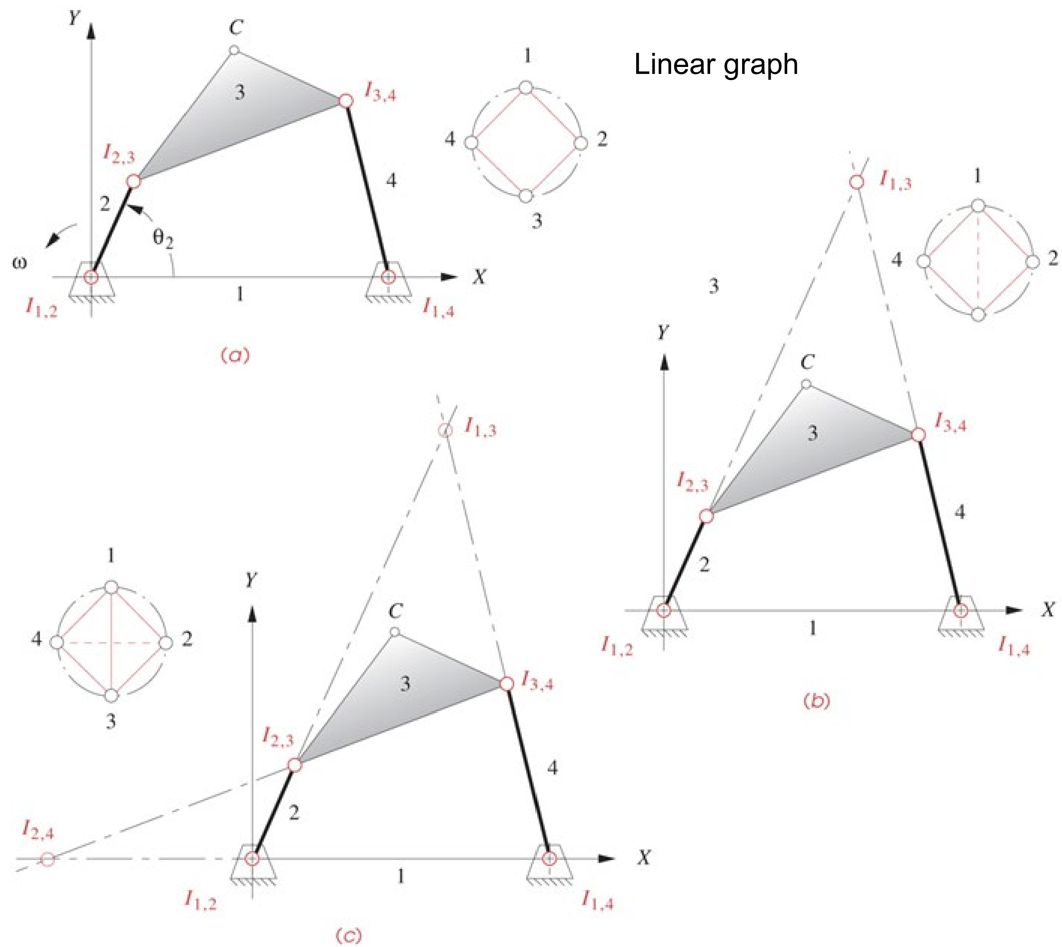
Instant centre (IC) of velocity

A point common to two bodies which has the same velocity. In other words, the relative velocity of this point is ZERO.

Kennedy's rule: any three bodies in plane motion have three ICs which lie on a straight line.

Slider joint: IC is at infinity along a line perpendicular to the sliding direction.

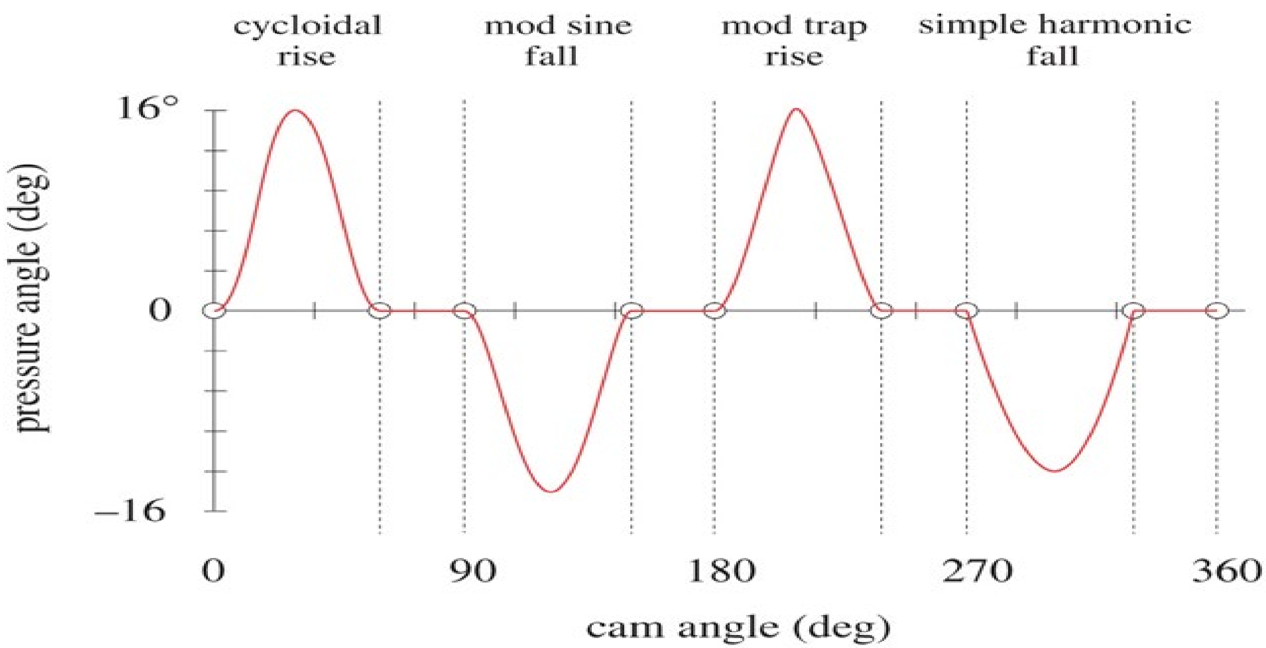




Prime circle radius and eccentricity selection based on

$$\phi = \arctan \frac{v - \varepsilon}{s + \sqrt{R_P^2 - \varepsilon^2}}$$

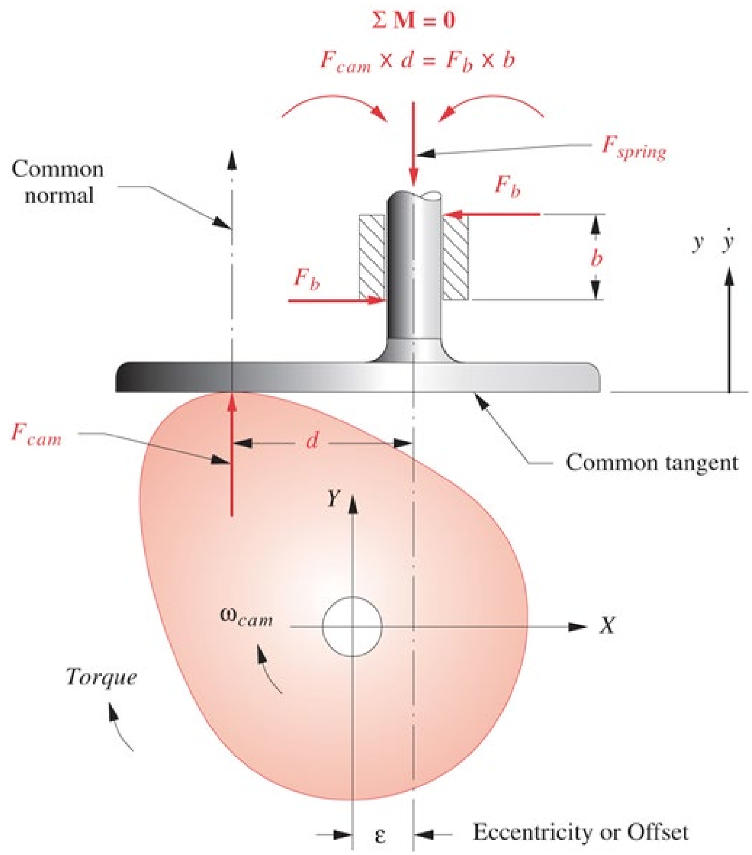
- Prime circle increases, pressure angle decreases
- Eccentricity increases, press angle decreases in the rise and increases in the fall.



One example: pressure angle function similar to velocity

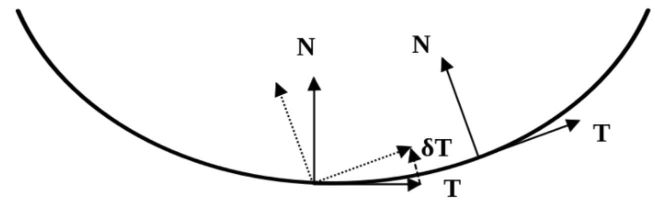
Overturning moment (jamming)- flat-faced follower (pressure angle =0)

- In this case, too large moment is caused by the force from the cam
- One solution is too keep the cam small in size to reduce the moment arm of the force (*d*)
- Eccentricity affects the average value of the moment.



Radius of curvature - roller follower

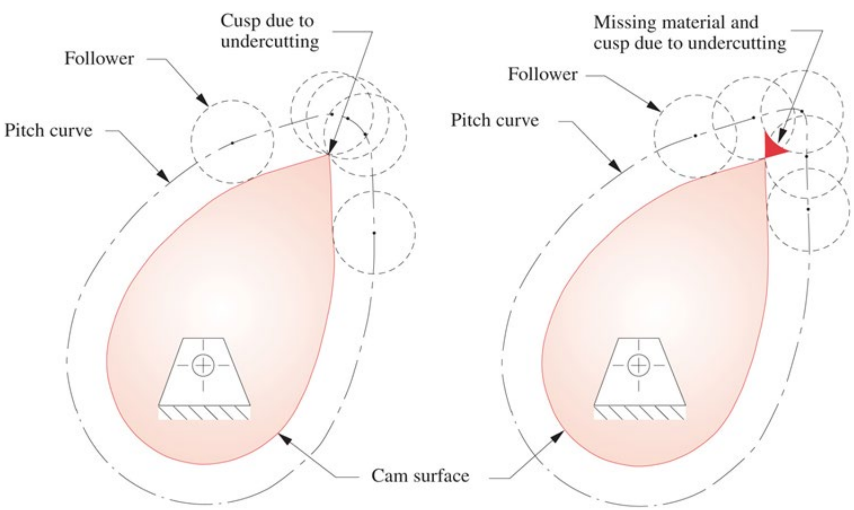
$$\rho_{pitch} = \frac{\left[(R_P + s)^2 + v^2\right]^{3/2}}{(R_P + s)^2 + 2v^2 - a(R_P + s)}$$



Curvature $\quad k = \left\| \frac{dT}{ds} \right\|$

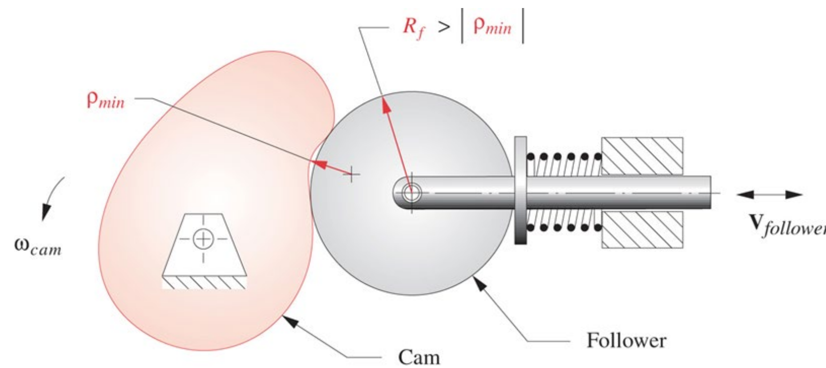
Radius of Curvature $\quad \rho = {}^{1}/_{k}$

Undercutting (cusp):



(a) Radius of curvature of pitch curve equals the radius of the roller follower

(b) Radius of curvature of pitch curve is less than the radius of the roller follower



The rule of thumb: $\quad |\rho_{min}| >> R_f$

Radius of curvature – flat-faced follower

$$c\cos(\theta+\alpha) = x$$

$$c\sin(\theta+\alpha) + \rho = R_b + s$$

Differentiate them w.r.t to θ

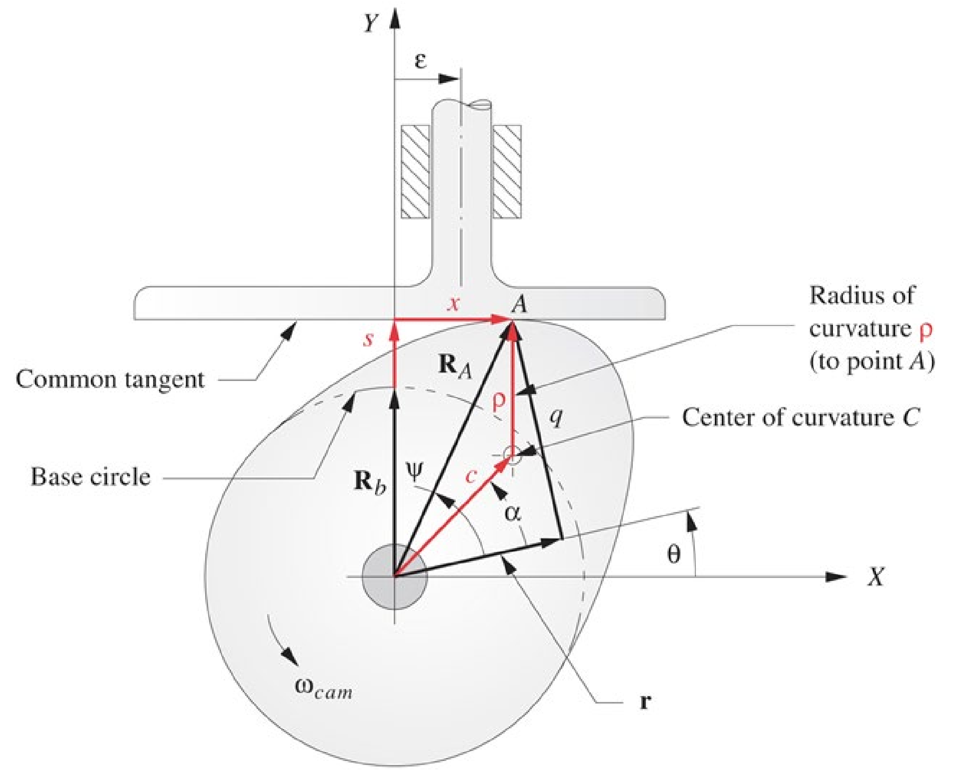
$$-c\sin(\theta+\alpha) = \frac{dx}{d\theta}$$

$$c\cos(\theta+\alpha) = \frac{ds}{d\theta} = v = x$$

$x = v$, $\underline{facewidth > v_{max} - v_{min}}$

$$\frac{dx}{d\theta} = \frac{dv}{d\theta} = a \longrightarrow \underline{\rho = R_b + s + a}$$

$$\rho_{min} = R_b + (s+a)_{min}$$



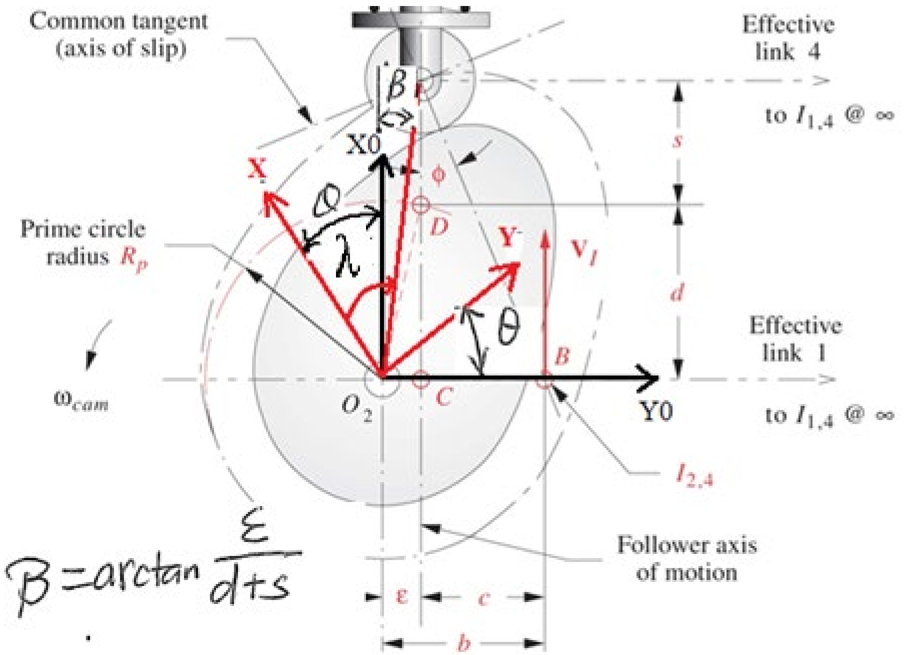
Contour of the cam (locus of the roller centre) - roller follower

After the prime circle radius and base circle radius are determined for suitable pressure and minimum radius of curvature, the contour of the cam can be defined.

$$x = \cos\lambda\sqrt{(d+s)^2 + \varepsilon^2}$$

$$y = \sin\lambda\sqrt{(d+s)^2 + \varepsilon^2}$$

$$\lambda = \theta + \arctan\left(\frac{\varepsilon}{d+s}\right)$$



$$\beta = \arctan\frac{\varepsilon}{d+s}$$

$$\lambda = \beta + \theta = \theta + \arctan\frac{\varepsilon}{d+s}$$

Coordinates in the body frame

Contour of the cam – flat-faced follower

Base circle : $R_{b_{\min}} > \rho_{\min} - (s+a)_{\min}$

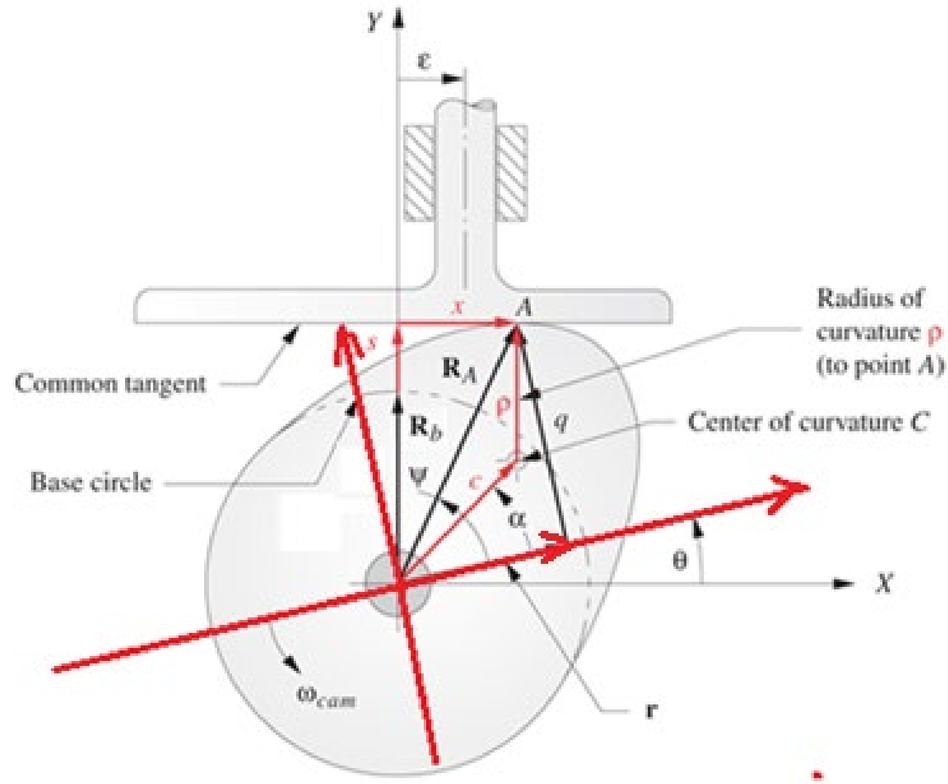
$$\mathbf{R}_A = x + j(R_b + s) = re^{j\theta} + qe^{j\left(\theta + \frac{\pi}{2}\right)}$$

$$r + jq = xe^{-j\theta} + j(R_b + s)e^{-j\theta}$$

$$x = v$$

$$r = (R_b + s)\sin\theta + v\cos\theta$$

$$q = (R_b + s)\cos\theta - v\sin\theta$$



Coordinates in the body frame